

Natural Language Processing

Decoding Strategies and Evaluations for Natural Language Generation

Outline

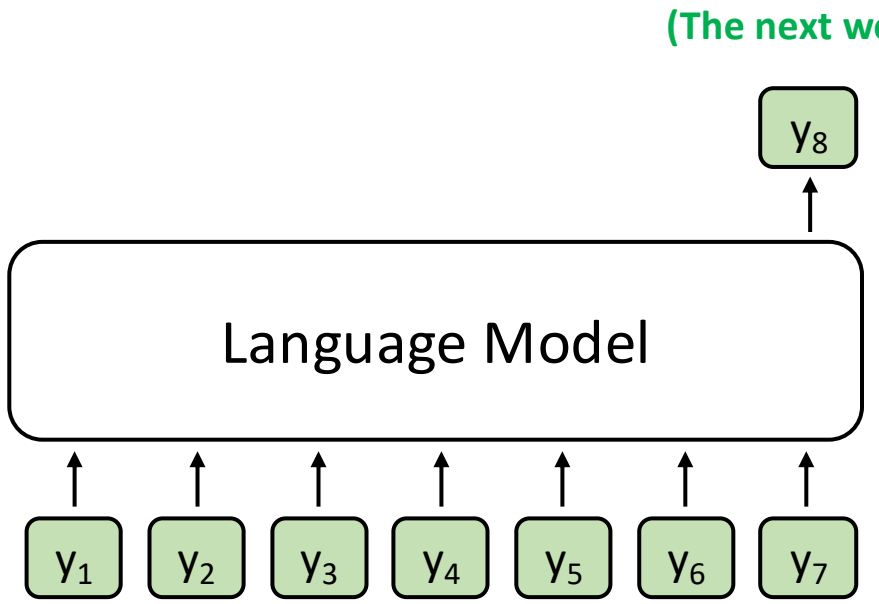
- Recap: Language Generation
- Decoding Strategies
  - Greedy Decoding
  - Beam Search
  - Top-k / Top-p Sampling
- Evaluations

Natural Language Generation (NLG)

- Natural language generation (NLG) is a **process** that **outputs** text.
- NLG includes a wide variety of NLP tasks.

Machine Translation | Abstractive Summarization | Dialogue Generation (e.g., ChatGPT) | Story Generation | Image Captioning | ...

Recap: Language Model



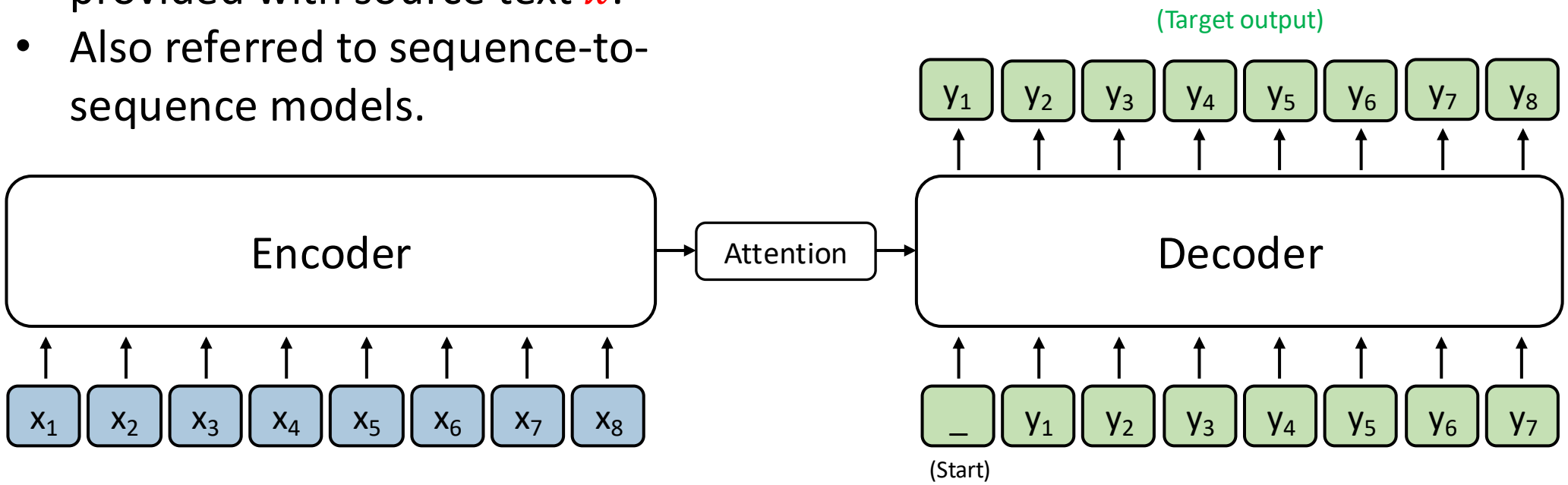
$$P(y_t|y_1, y_2, \cdots, y_{t-1})$$

- A model that assigns probabilities to upcoming words is called **a language model**.
- The task involving predictions of upcoming words is **language modeling**.

Recap: Conditional Language Model

- In addition to previous words, a conditional language model is provided with source text x.
- Also referred to sequence-to-sequence models.

$$P(y_t|y_1, y_2, \cdots, y_{t-1}, x)$$



Tasks of Conditional Language Model

- In addition to previous words (target), a conditional language model is provided with source text x.

	Source	Target
Machine Translation	Language A	Language B
Summarization	Long Text	Concise Text
Dialogue Generation	User Input	Desired User Input
…		

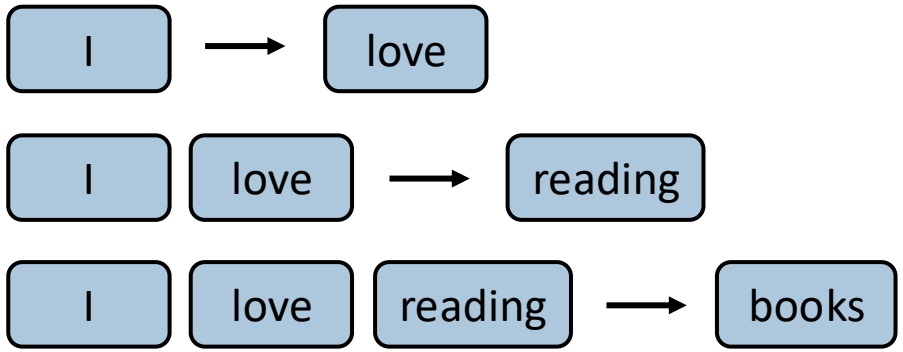
How to train a (Conditional) Language Model?

- First, you need a training (parallel) corpus.

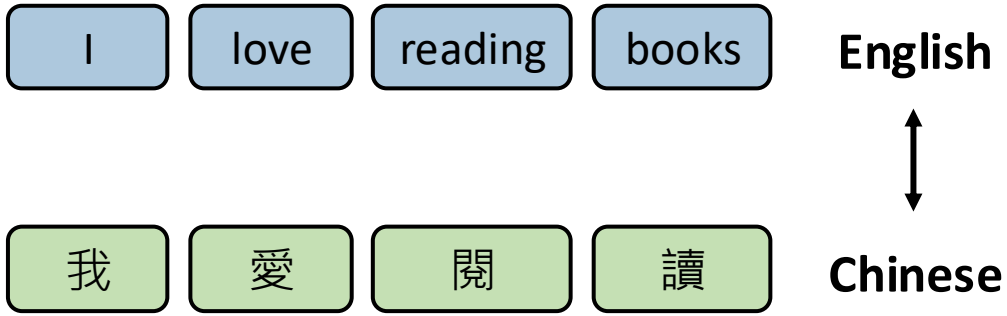
Supervised, Aligned

Example: I love reading books.

Language modeling (Unsupervised)

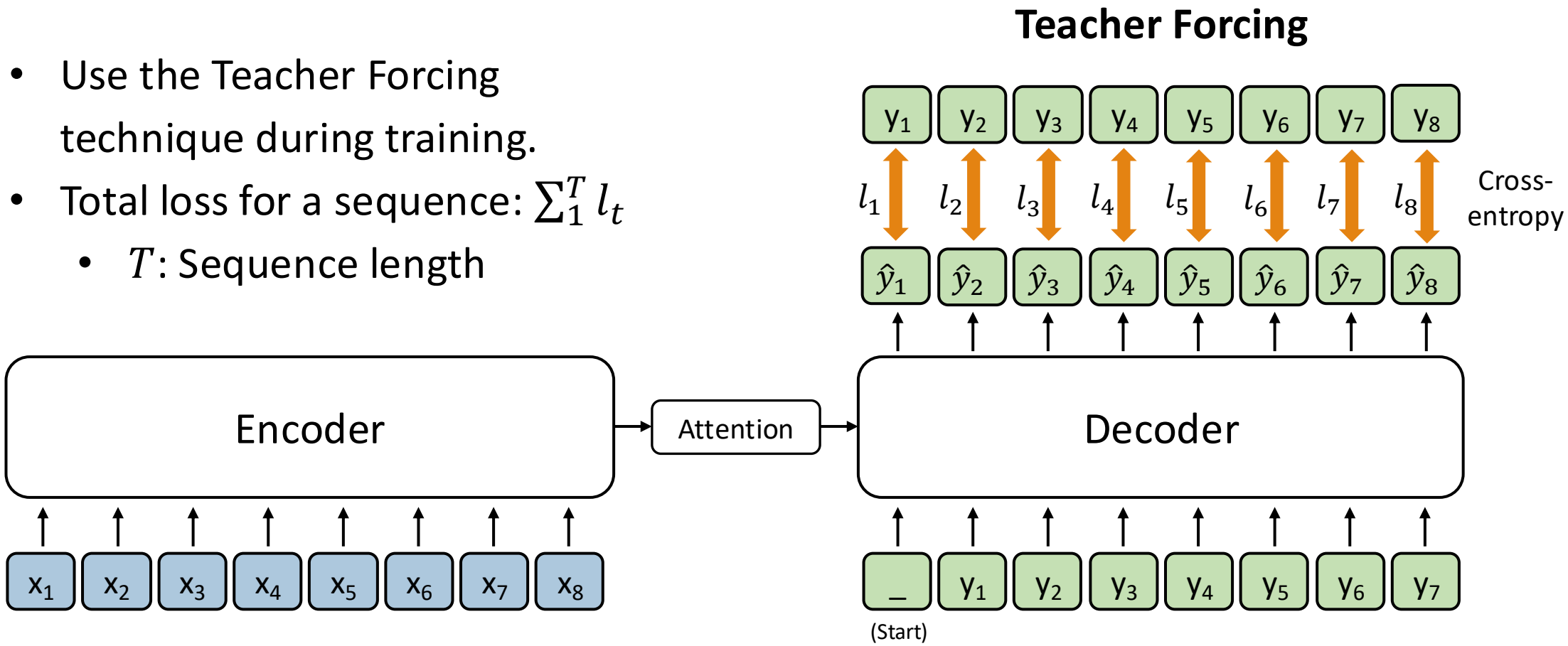


Machine Translation (Supervised)



How to train a (Conditional) Language Model?

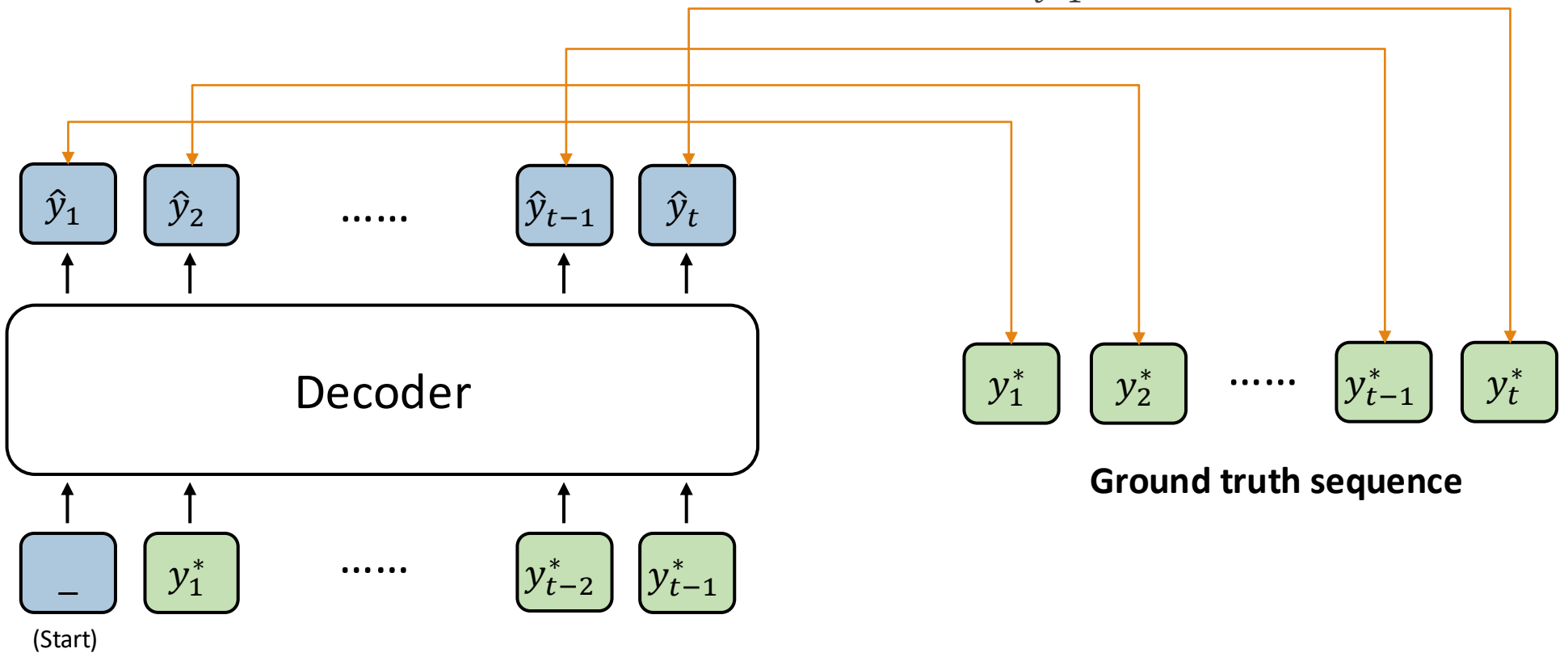
- Use the Teacher Forcing technique during training.
- Total loss for a sequence: $\sum_1^T l_t$
  - T: Sequence length



Teacher Forcing – Training stage

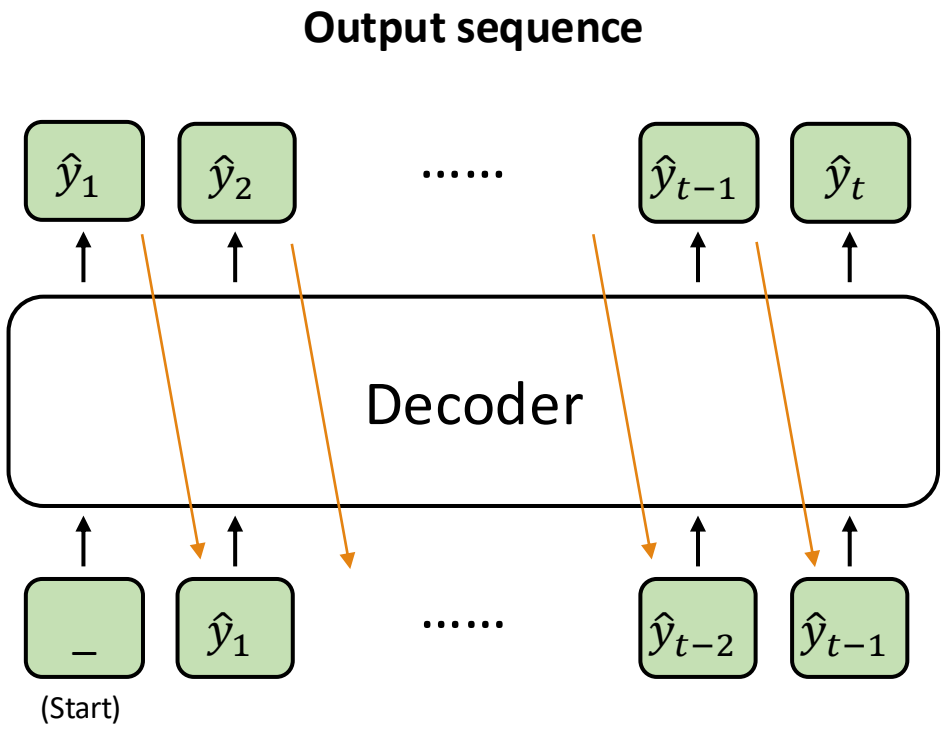
During training:

$$L_{ml} = -\sum_{t=1}^{n'} \log p(y_t^* | y_1^*, \ldots, y_{t-1}^*, x)$$



Teacher Forcing – Testing stage

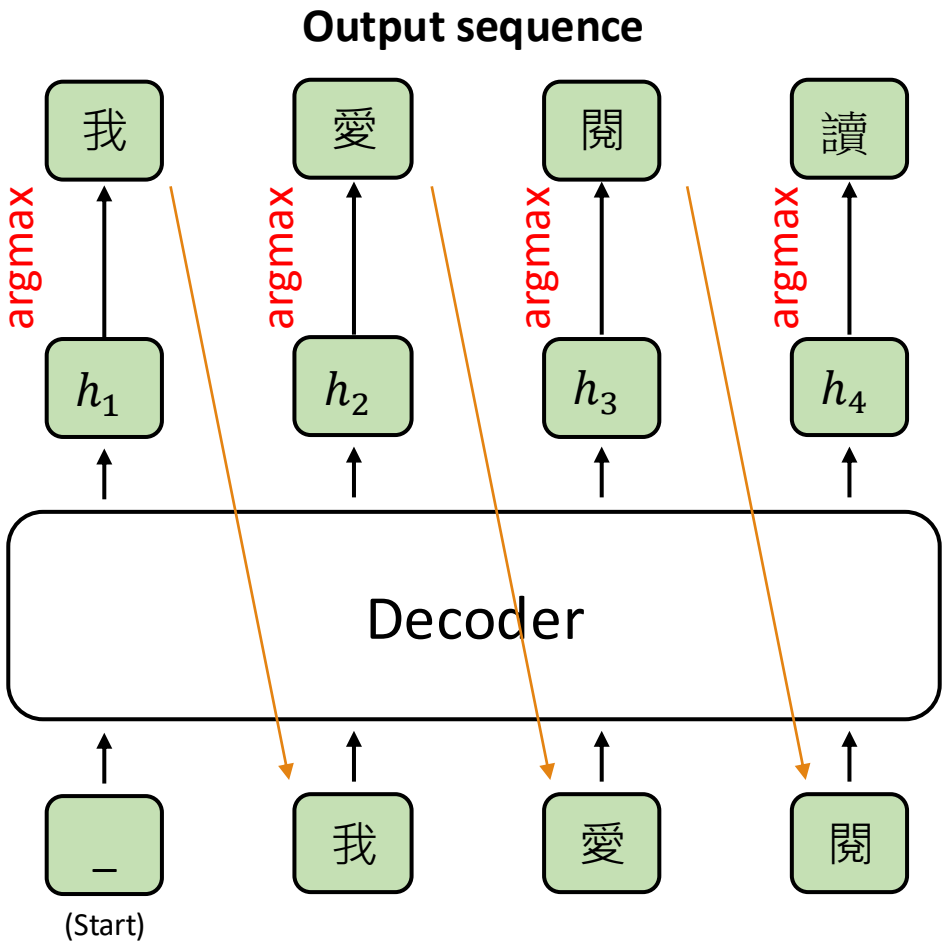
During testing:



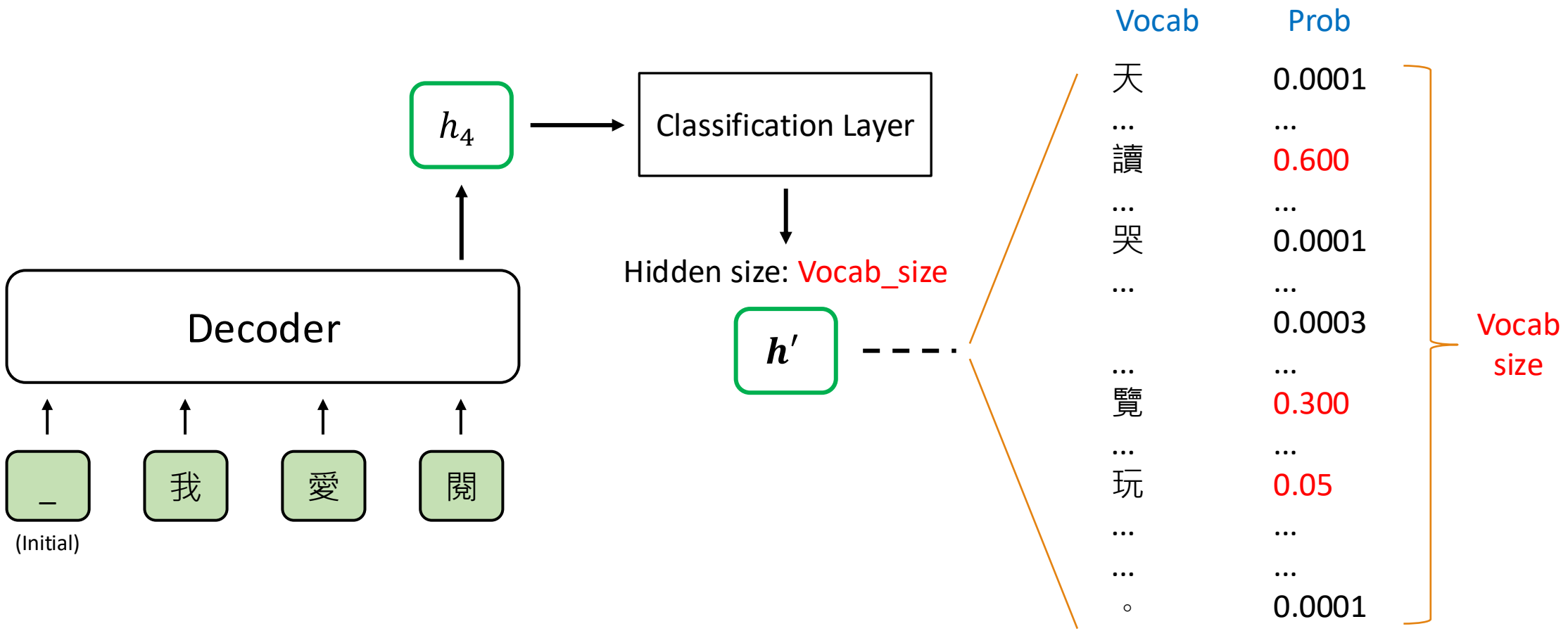
- Advantage: stabilize training and increase performance
- Question: How does the next word be determined?

Greedy Decoding

Example: I love reading books.

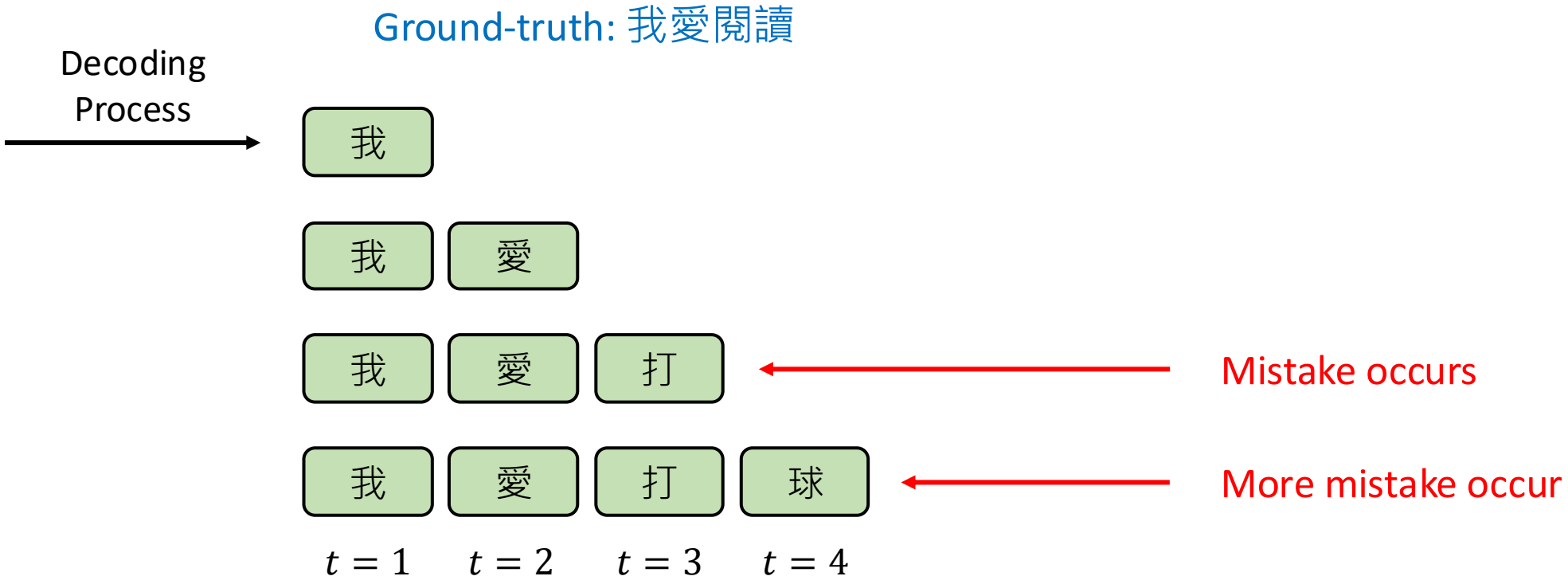


Greedy Decoding – Best Selection Process



Problem of Greedy Decoding

- Greedy decoding cannot undo!



Re-thinking Greedy Decoding

- Greedy decoding cannot undo!
- Greedy decoding only provides one best choice at each time step.
- How about providing more than one choice at each time step?
  - Stochastic Sampling (Top-k, Top-p sampling)
  - Deterministic Search → **Beam Search**
  - Top-p + Beam?

All applied on the last layer output (logits, e.g., 32k token probabilities)

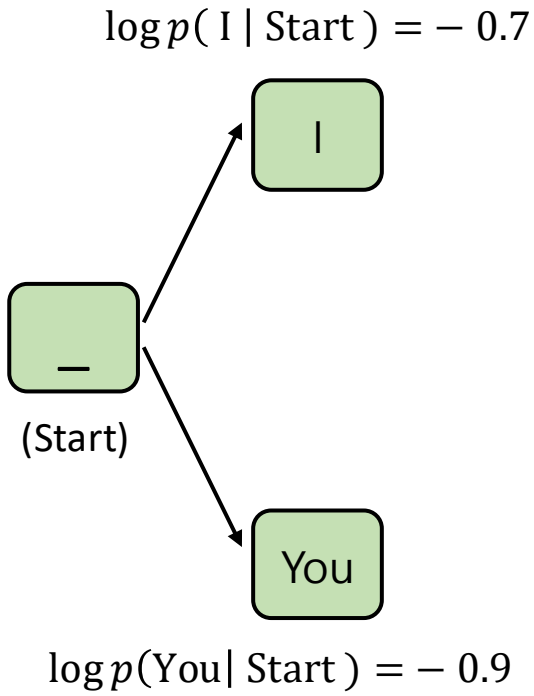
Beam Search

- Set the `Beam size` (or `Beam width`) = 2
  - This means that the number of candidates will be preserved at each decoding time.
  - Beam size is a hyperparameter for beam search decoding.
- At each decoding time step, a score is calculated via the following equation:

$$L_{ml} = \sum_{t=1}^{n'} \log p(y_t^* | y_1^*, \ldots, y_{t-1}^*, x)$$

Beam Search ($t = 1$)

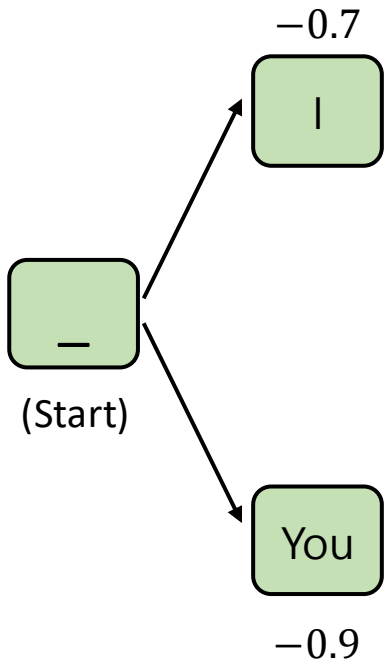
`Beam size` = 2



- At this decoding step, two choices are preserved.

Beam Search ($t = 1$)

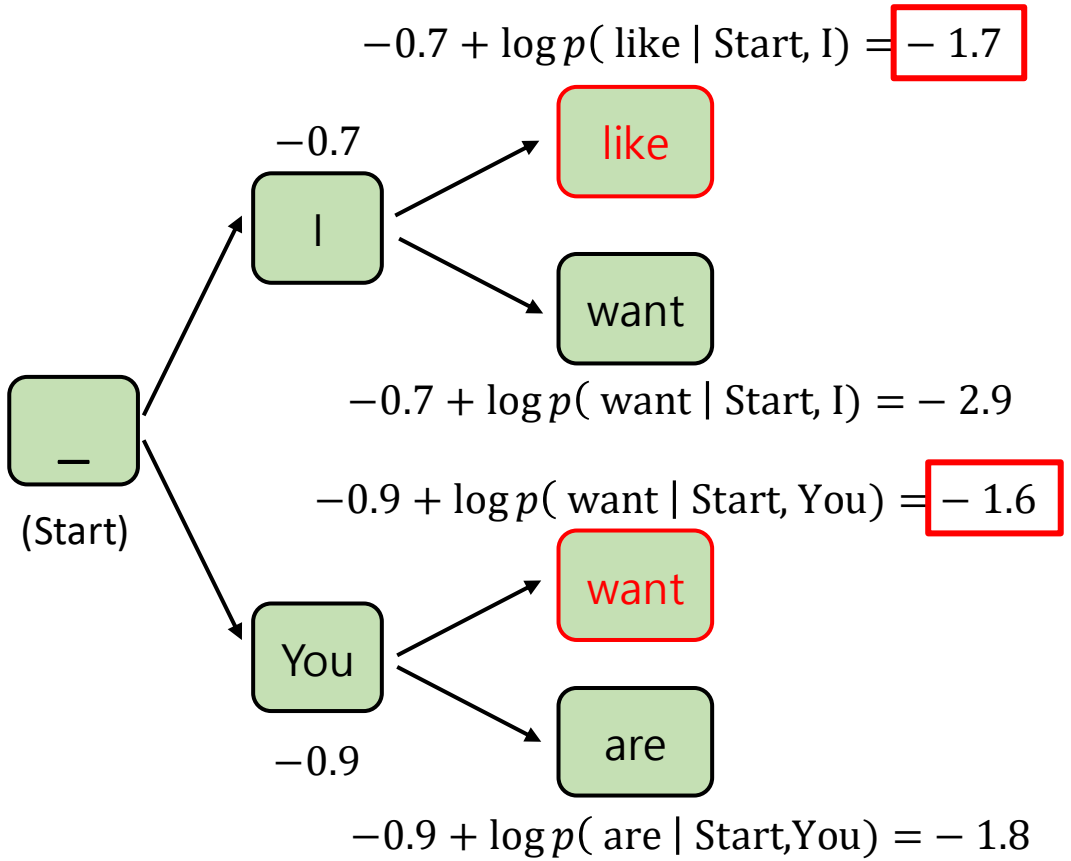
`Beam size` = 2



- At this decoding step, two choices are preserved.

Beam Search ($t = 2$)

`Beam size` = 2

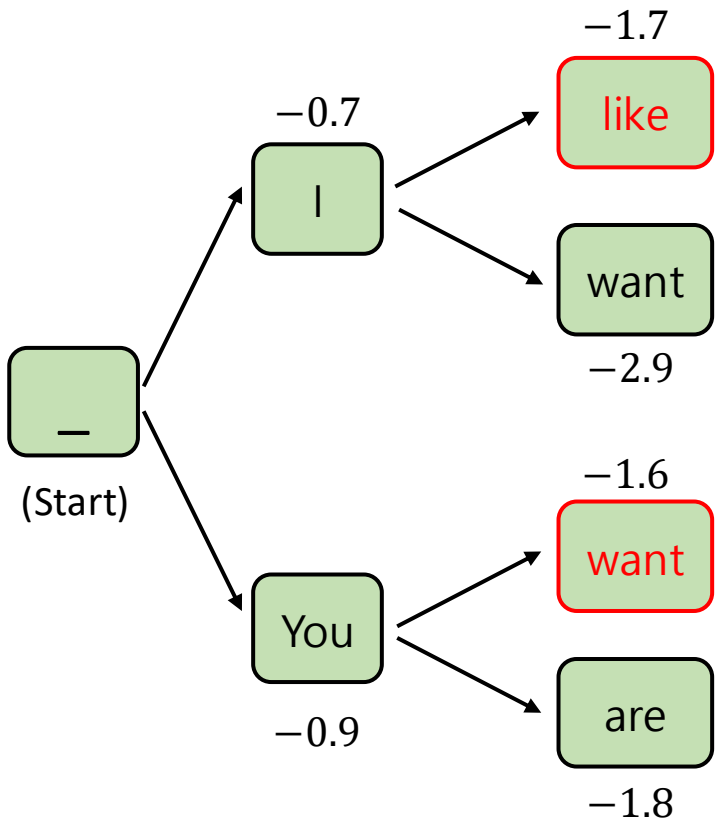


Note the loglikelihood! Being close to zero is better!

- At this decoding step, two choices are preserved, and the other two are discarded.

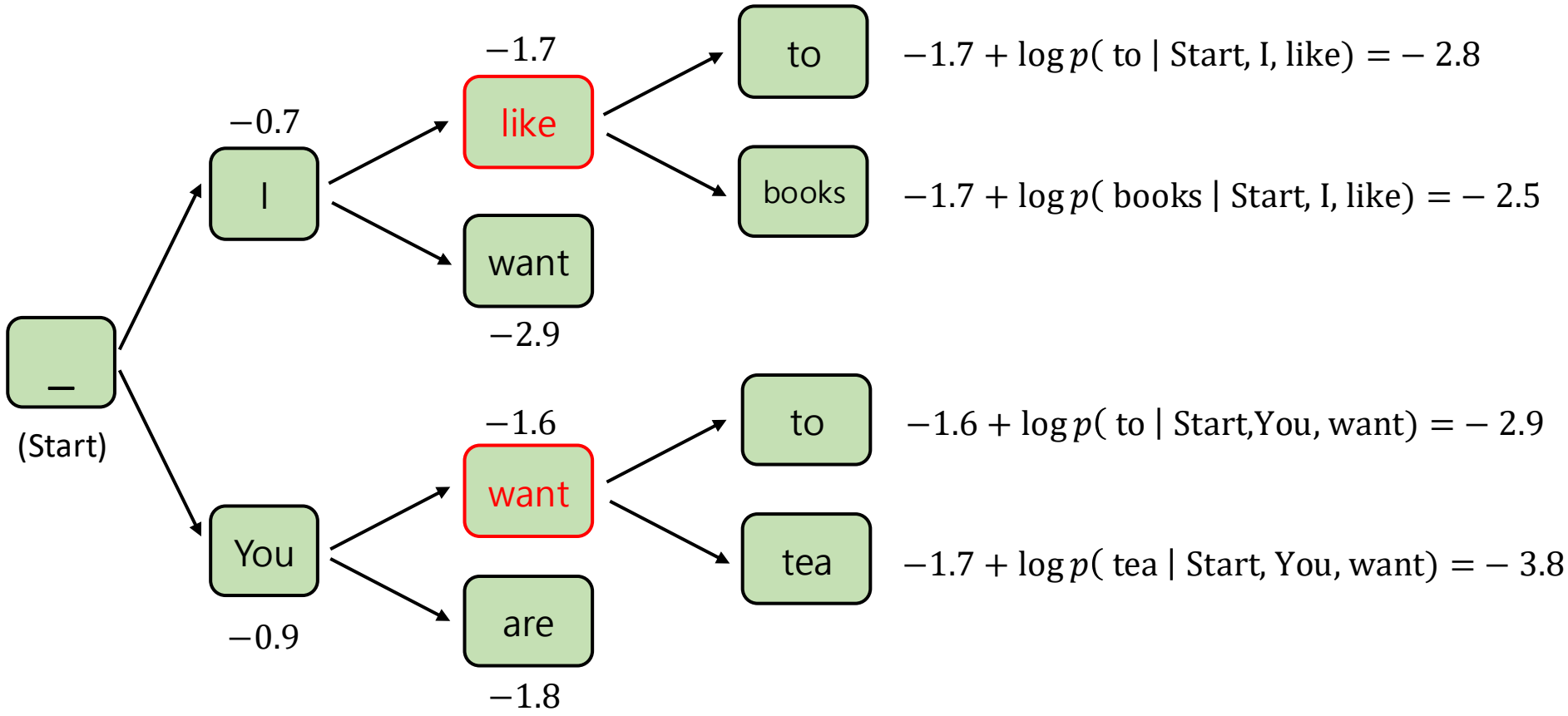
Beam Search ($t = 2$)

`Beam size` = 2



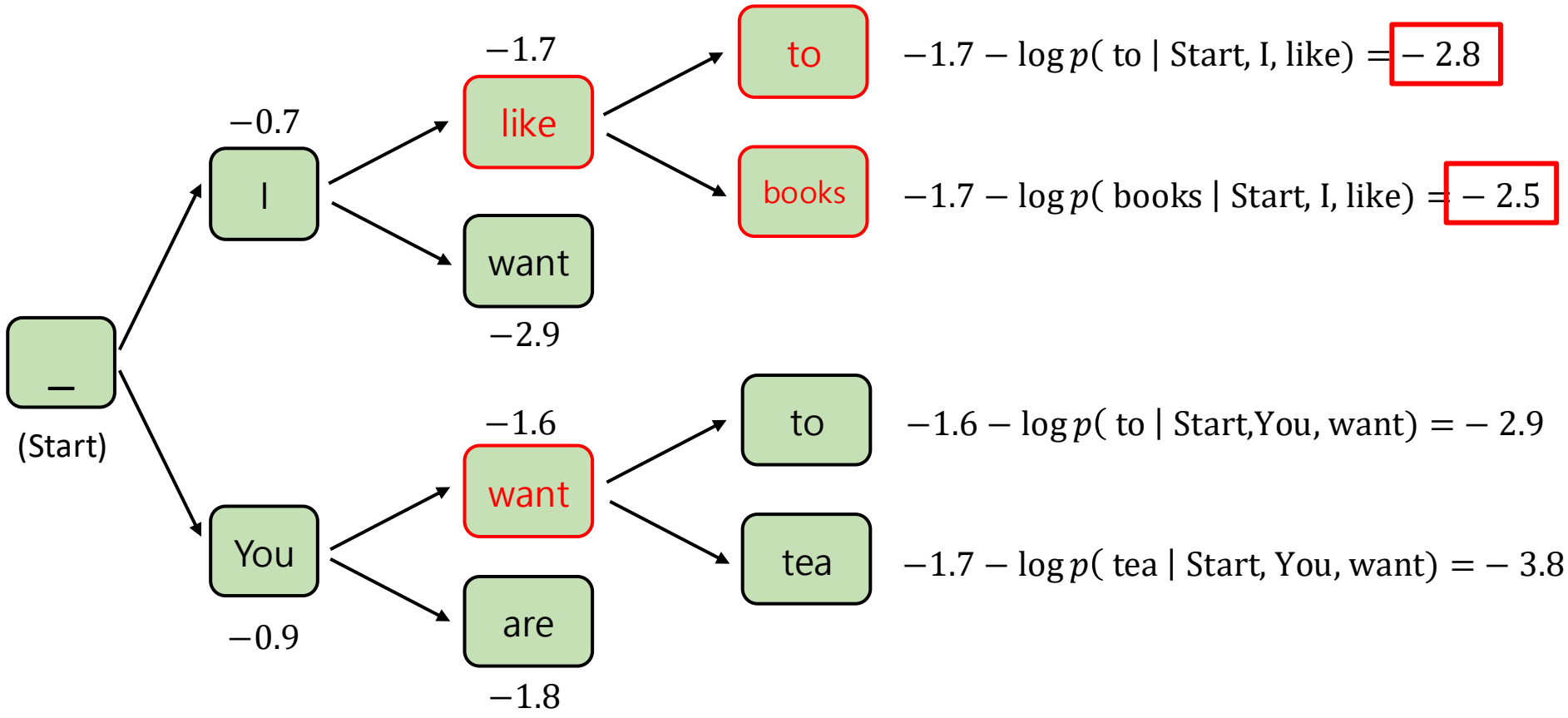
Beam Search ($t = 3$)

`Beam size` = 2



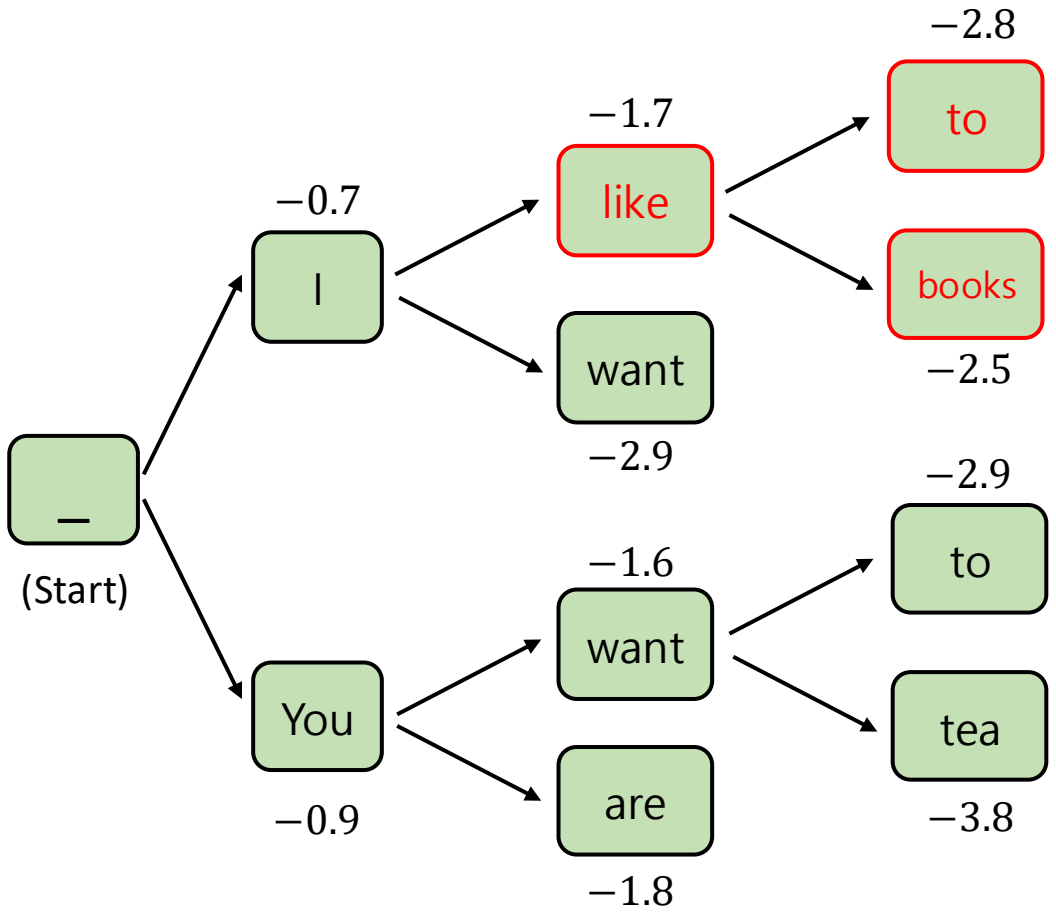
Beam Search ($t = 3$)

`Beam size` = 2



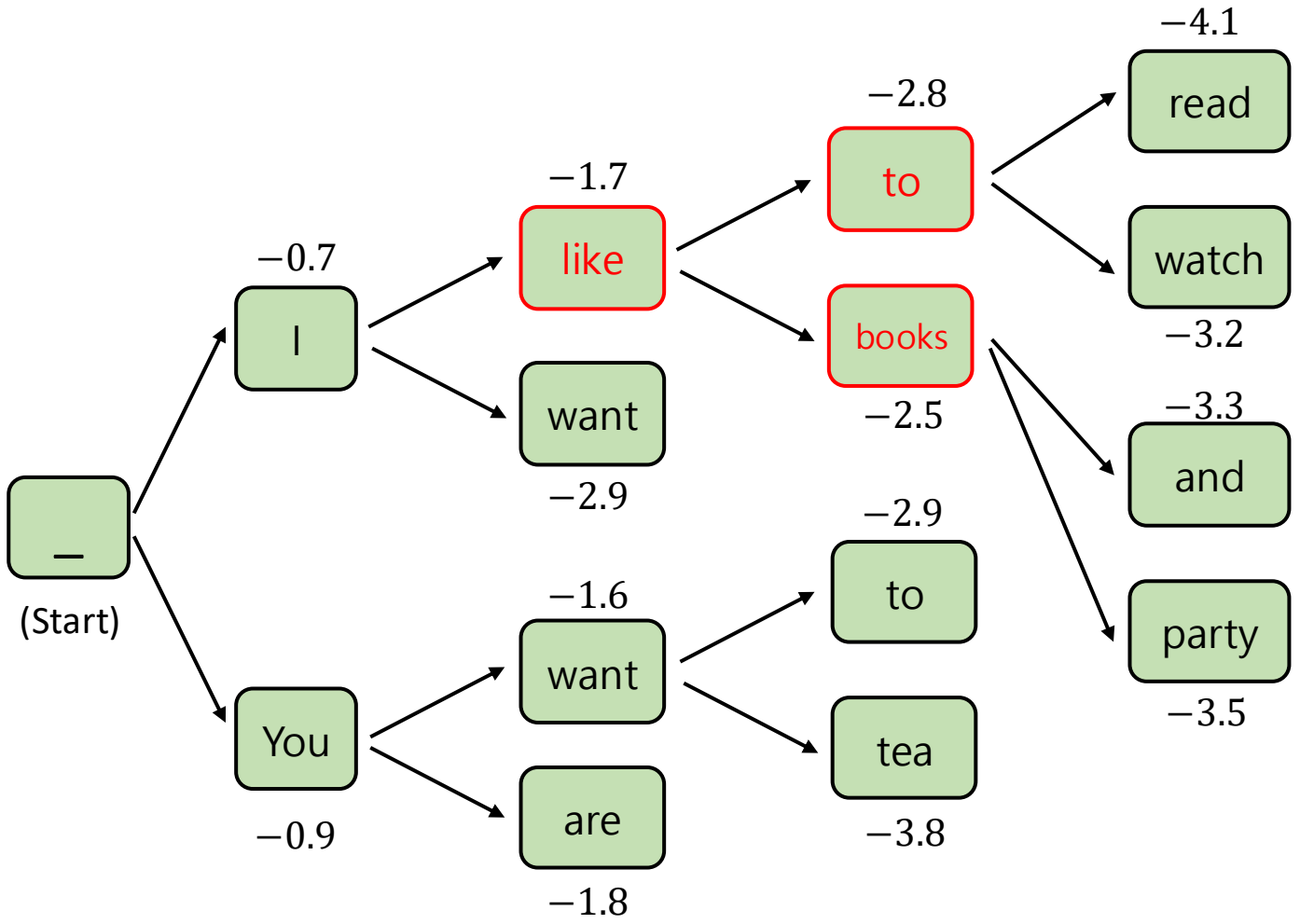
Beam Search ($t = 3$)

`Beam size` = 2



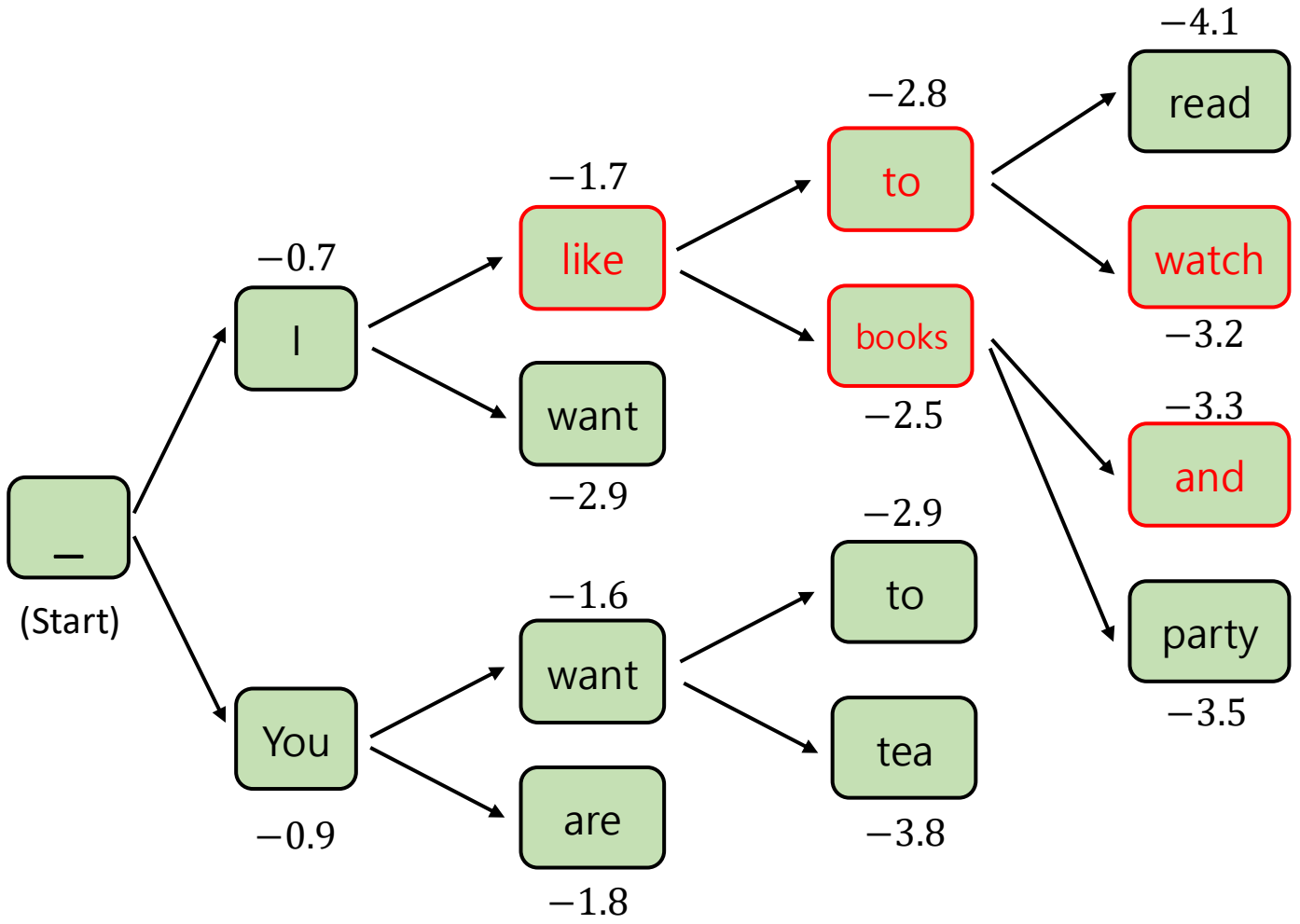
Beam Search ($t = 4$)

`Beam size` = 2



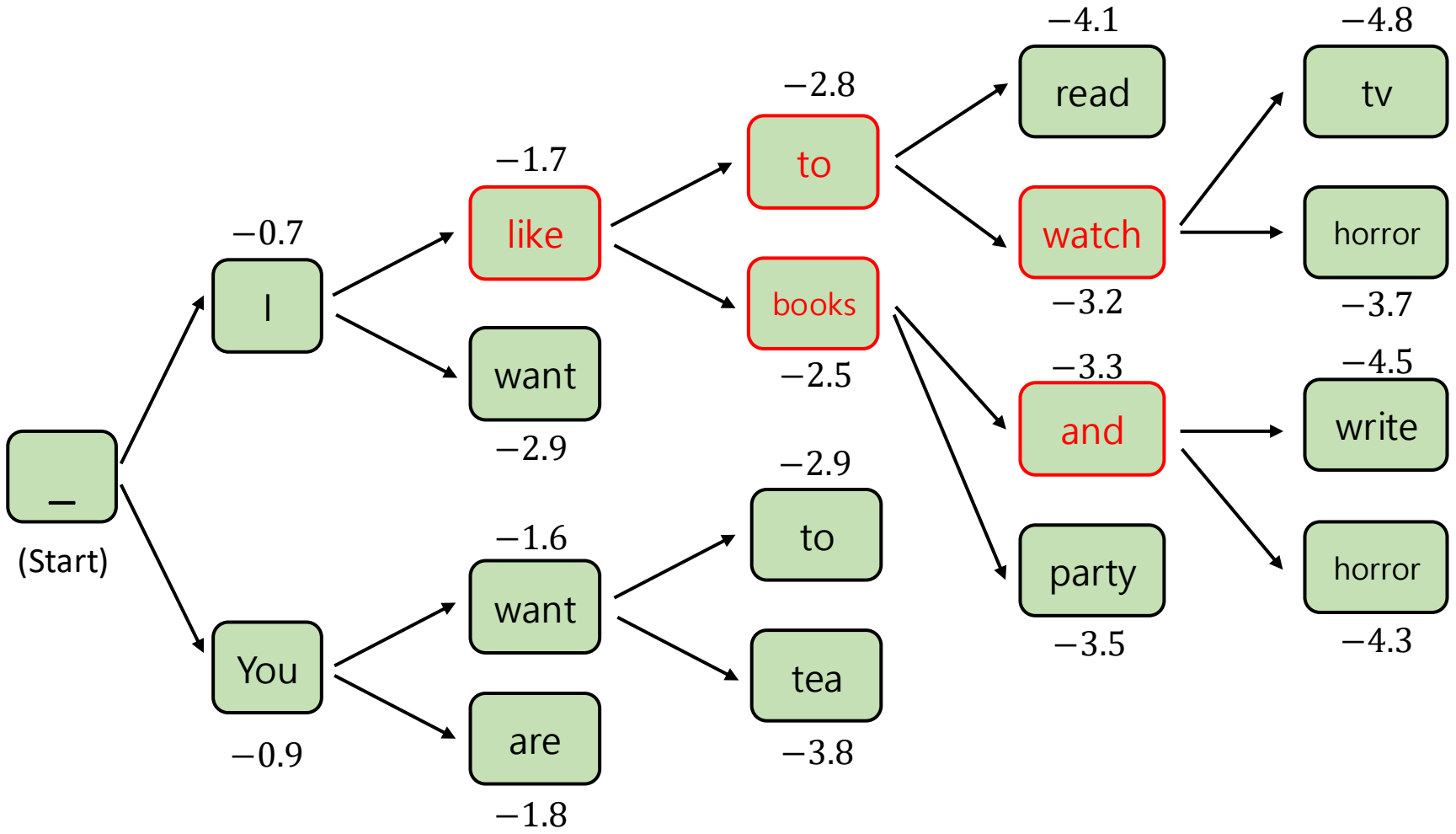
Beam Search ($t = 4$)

`Beam size` = 2



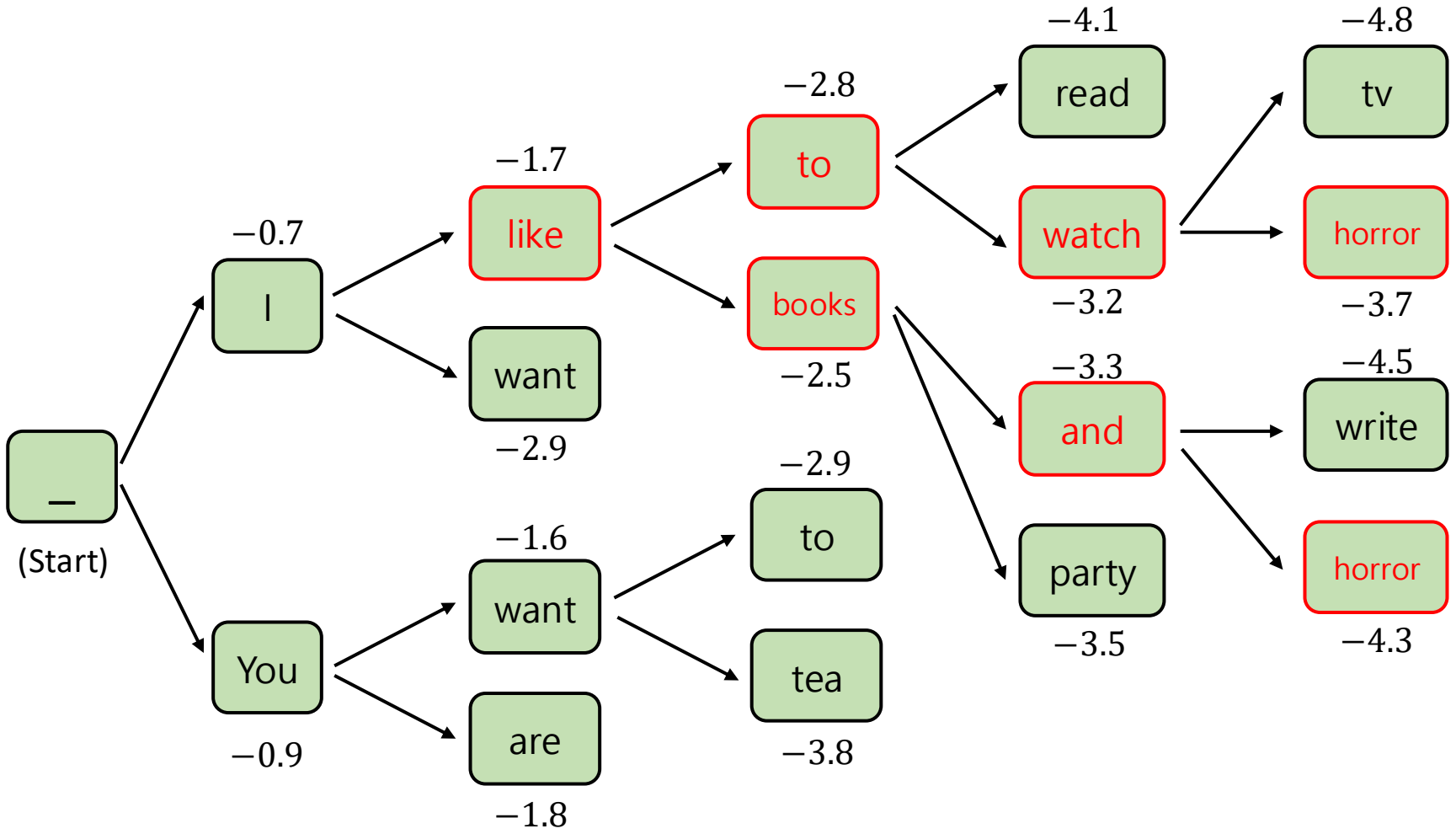
Beam Search ($t = 5$)

`Beam size` = 2



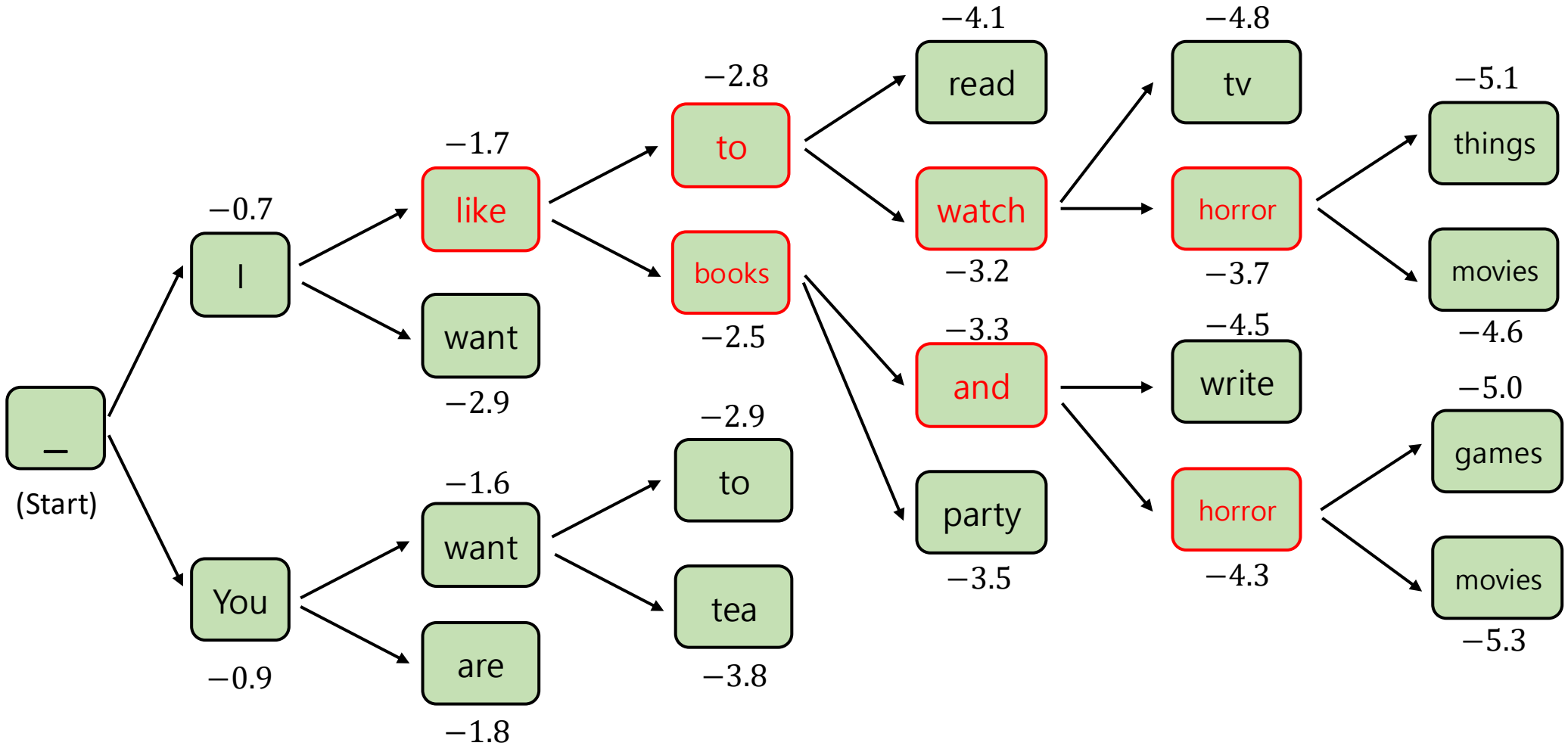
Beam Search ($t = 5$)

`Beam size` = 2



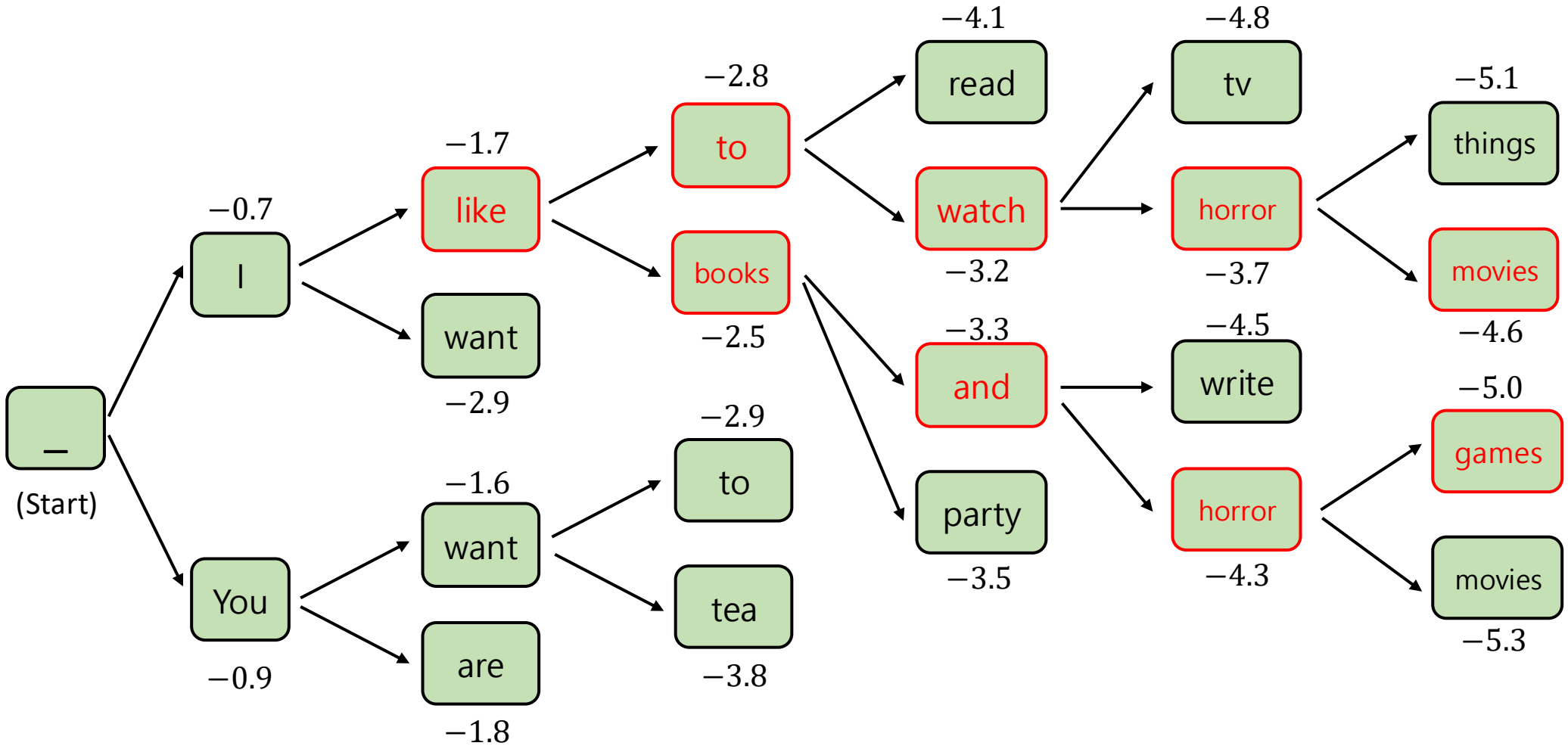
Beam Search ($t = 6$)

`Beam size` = 2



Beam Search ($t = 6$)

`Beam size` = 2



Stop Criterion

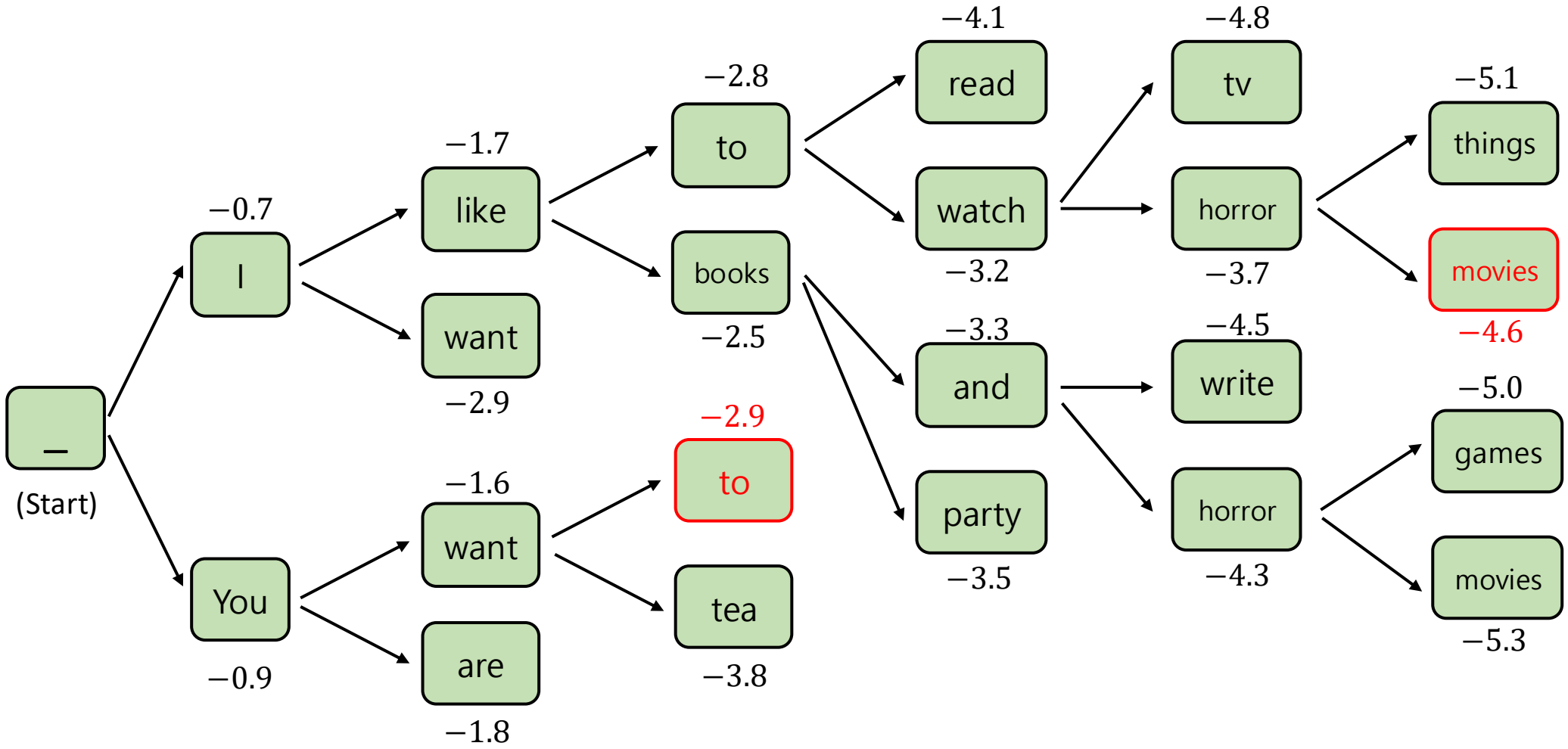
- There are two common stop criteria, either for greedy decoding or beam search decoding:
  - We consider a sequence of generation complete when the <EOS> token is produced by a model. *<EOS>: End of sequence
    - E.g., <Start> I like to watch horror movies <EOS>
  - A generated sequence reaches a pre-defined maximal length.

Problem of Beam Search

- Longer candidates will have lower scores.
  - (Let's see again the 6th time step)
- lead to text that is **degenerated**—it is **repetitive, generic, and unnatural**
  - The Curious Case of Neural Text Degeneration (https://arxiv.org/abs/1904.09751 , ICLR 2020)
  - Top-P
- High cost (B x KV-cache, low throughput)

Beam Search ($t = 6$)

`Beam size` = 2



Problem of Beam Search

- Longer candidates will have lower scores.
- Solution: Perform normalization to penalize on length

$$L_{ml} = \frac{1}{T}\sum_{t=1}^{T} \log p(y_t^* | y_1^*, \ldots, y_{t-1}^*, x)$$

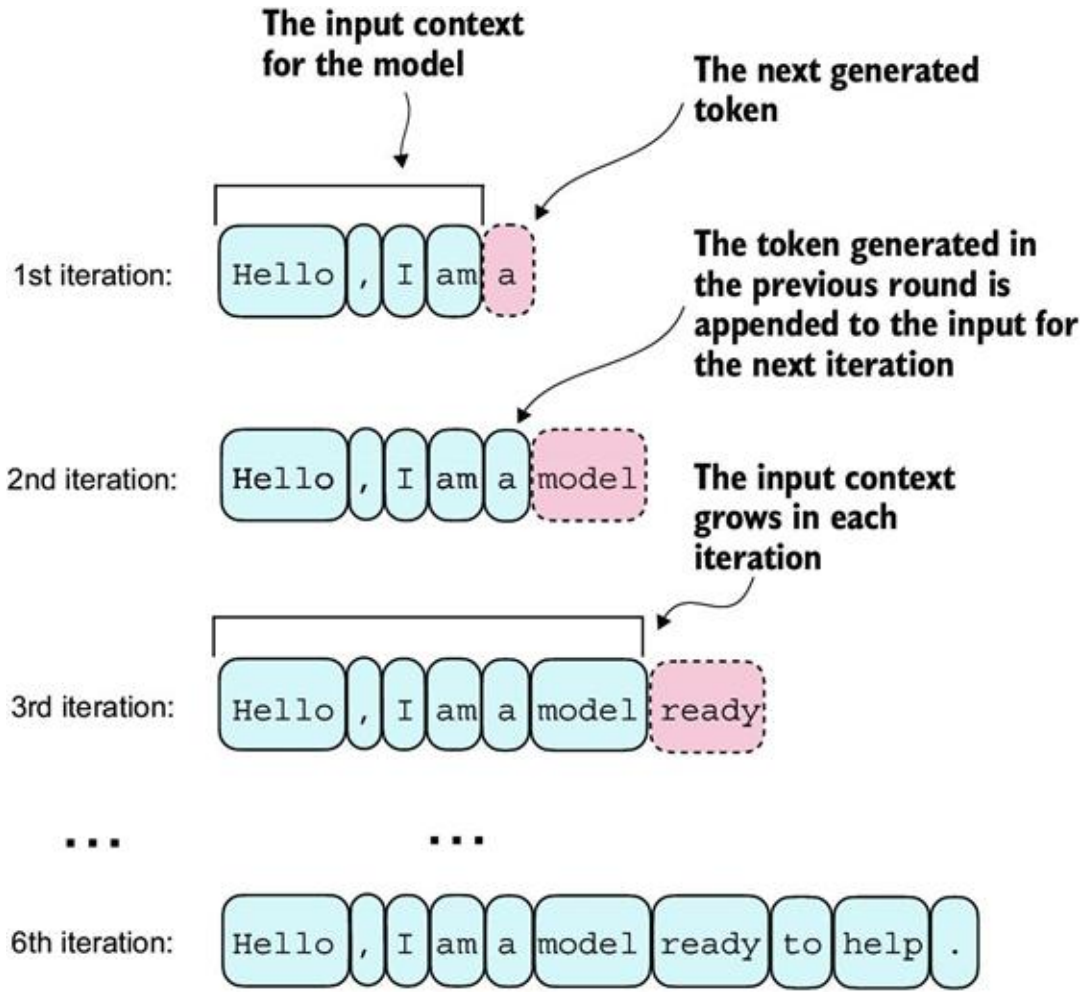
ACCELERATING LARGE LANGUAGE MODEL INFERENCE WITH DRAFT-AND-VERIFY

Speculative Decoding

The Problem: Autoregressive Decoding

LLMs generate tokens one at a time, with each new token depending on the preceding context.

- **Sequential Dependency**
  - Each new token depends on the previously generated context.
- **Repeated Inference**
  - Generating N tokens typically requires approximately N sequential decoding
- **Latency Bottleneck**
  - Longer outputs require more sequential decoding steps, limiting generation speed even on highly parallel GPUs.

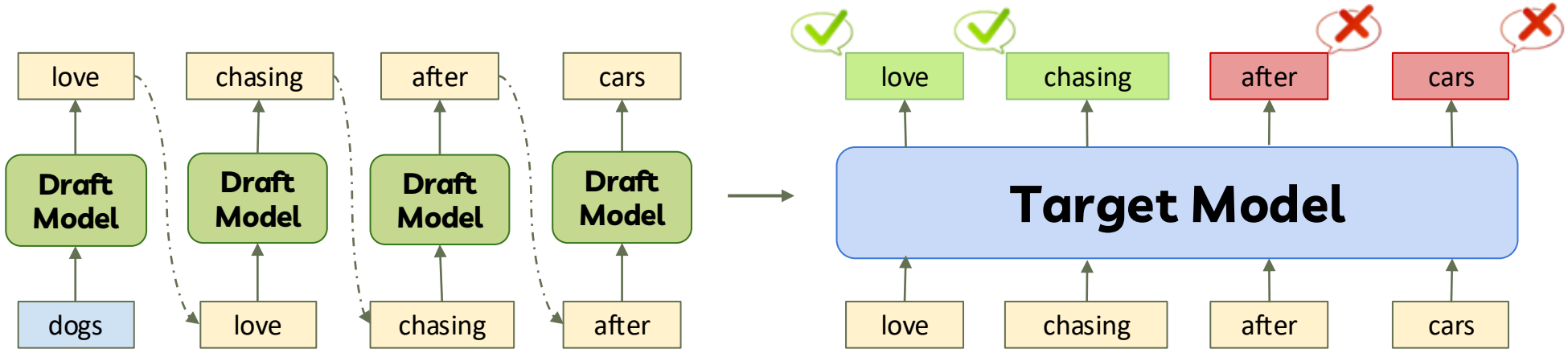


Speculative Decoding: Draft → Verify

The Key Idea: Predict First, Verify Later

1. Draft — A small model generates multiple candidate tokens.
2. Verify — The target model verifies these tokens in one forward pass.
3. Accept / Reject — Accept correct predictions and regenerate from the first rejected token.

Generate candidates sequentially with a lightweight model → Verify them in parallel with the target model.

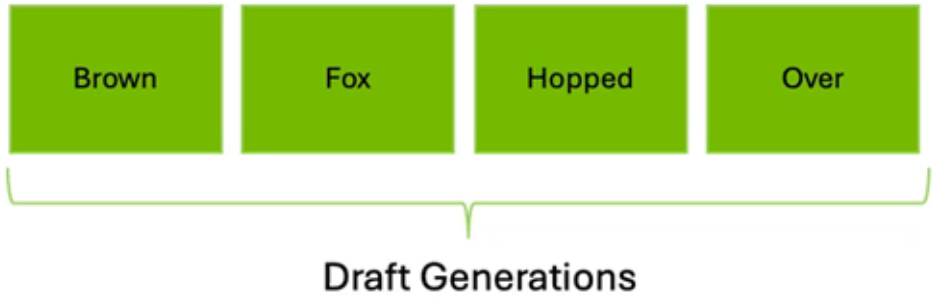


Step1. Autoregressive Generation

Step2. Parallel Verification

Accept or Reject: How Are Draft Tokens Verified?

1. Accept — Keep the proposed token according to the acceptance rule.
2. Reject — Reject a proposal according to the acceptance rule and sample a correction from the target distribution.



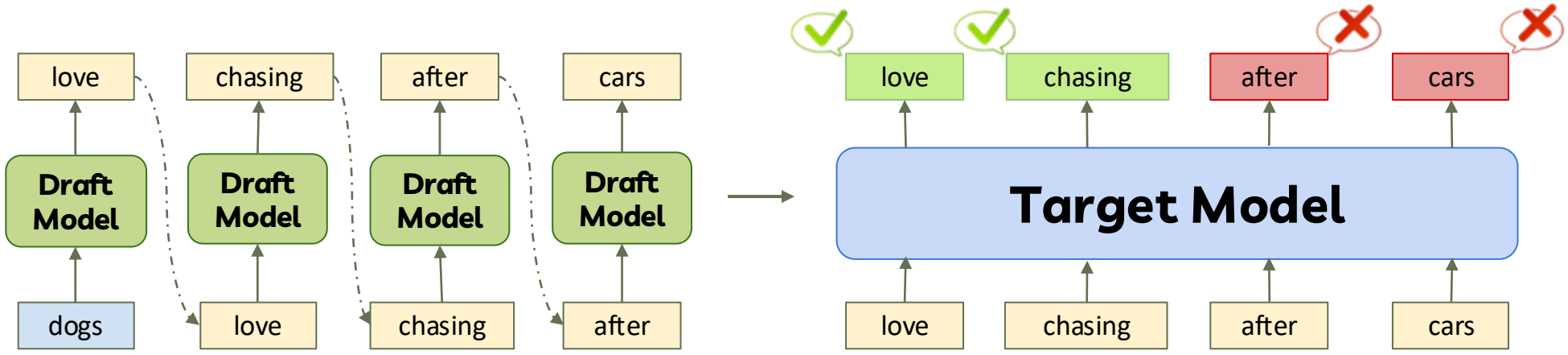
https://developer.nvidia.com/blog/an-introduction-to-speculative-decoding-for-reducing-latency-in-ai-inference/

Why Does It Work?

Generation is Sequential, but Verification is Parallelizable

- Drafting requires autoregressive generation.
- But once candidate tokens are available, the Target Model can evaluate multiple positions in one forward pass.
- Therefore, several expensive Target Model decoding steps can potentially be replaced by one verification step.

The speedup depends on whether the cost of drafting is smaller than the Target Model computation it replaces.



Step1. Autoregressive Generation **Step2. Parallel Verification**

Why Can Speculative Decoding Be Faster?

1. Draft with a Lightweight Model

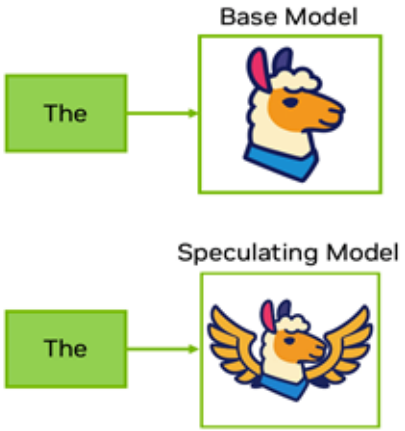
   The draft model sequentially generates N candidate tokens at a relatively low cost.

2. Verify in Parallel

   The target model evaluates multiple candidate positions in a single forward pass.

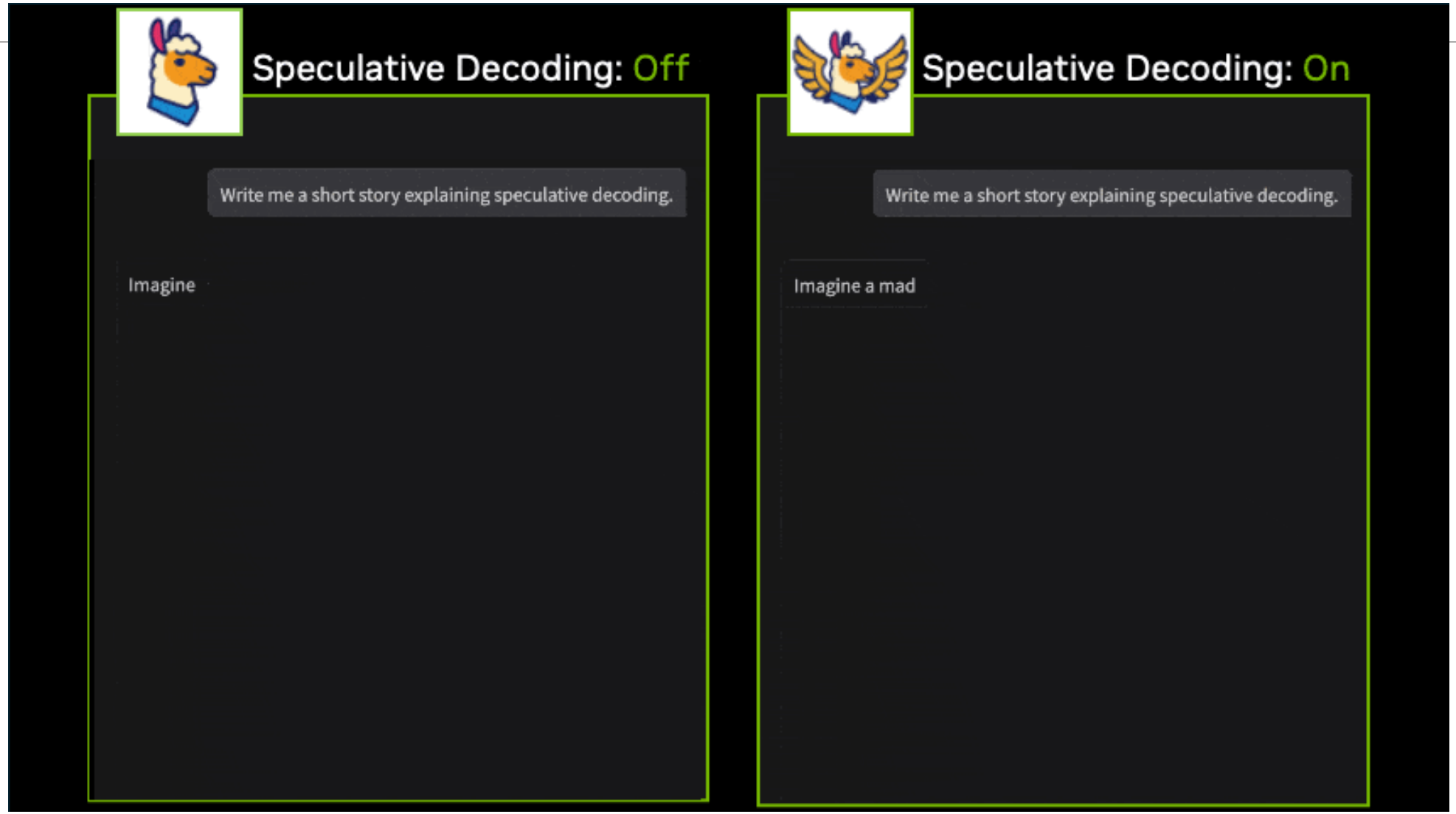
3. Reduce Expensive Decoding Steps

   If enough candidates are accepted, one verification cycle can replace several sequential target-model decoding steps.



https://developer.nvidia.com/blog/an-introduction-to-speculative-decoding-for-reducing-latency-in-ai-inference/

Speculative Decoding: Off vs. On



Two Directions of Speculative Decoding

Direction 1 — Better Drafting

Method	Key Idea
SpecInfer (2023)	Use a **token tree** to explore multiple candidate paths.
EAGLE (2024)	Use **hidden features** to build a lightweight drafter.
EAGLE-2 (2024)	Dynamically construct the draft tree according to context.
Dynamic Speculation (2023)	Dynamically adjust the number of draft tokens.

Direction 2 — Eliminate the Separate Drafter

Method	Key Idea
Medusa (2024)	Add multiple decoding heads to the target model for future-token prediction.
MTP (2024)	Integrate multi-token prediction directly into the model architecture.

EAGLE (Extrapolation Algorithm for Greater Language-model Efficiency)
Speculative Sampling Requires Rethinking Feature Uncertainty

Motivation: The Draft Model Is Itself a Bottleneck

Traditional speculative decoding relies on a smaller draft model: **Low drafting cost ↔ High prediction accuracy**

- Big Draft Model : High prediction accuracy, but high drafting cost and low efficiency.
- Small Draft Model: High efficiency with low drafting cost but low accuracy.

EAGLE's Key Insight

- **Feature-level** autoregression is easier than token-level autoregression.
- Feature prediction suffers from uncertainty.

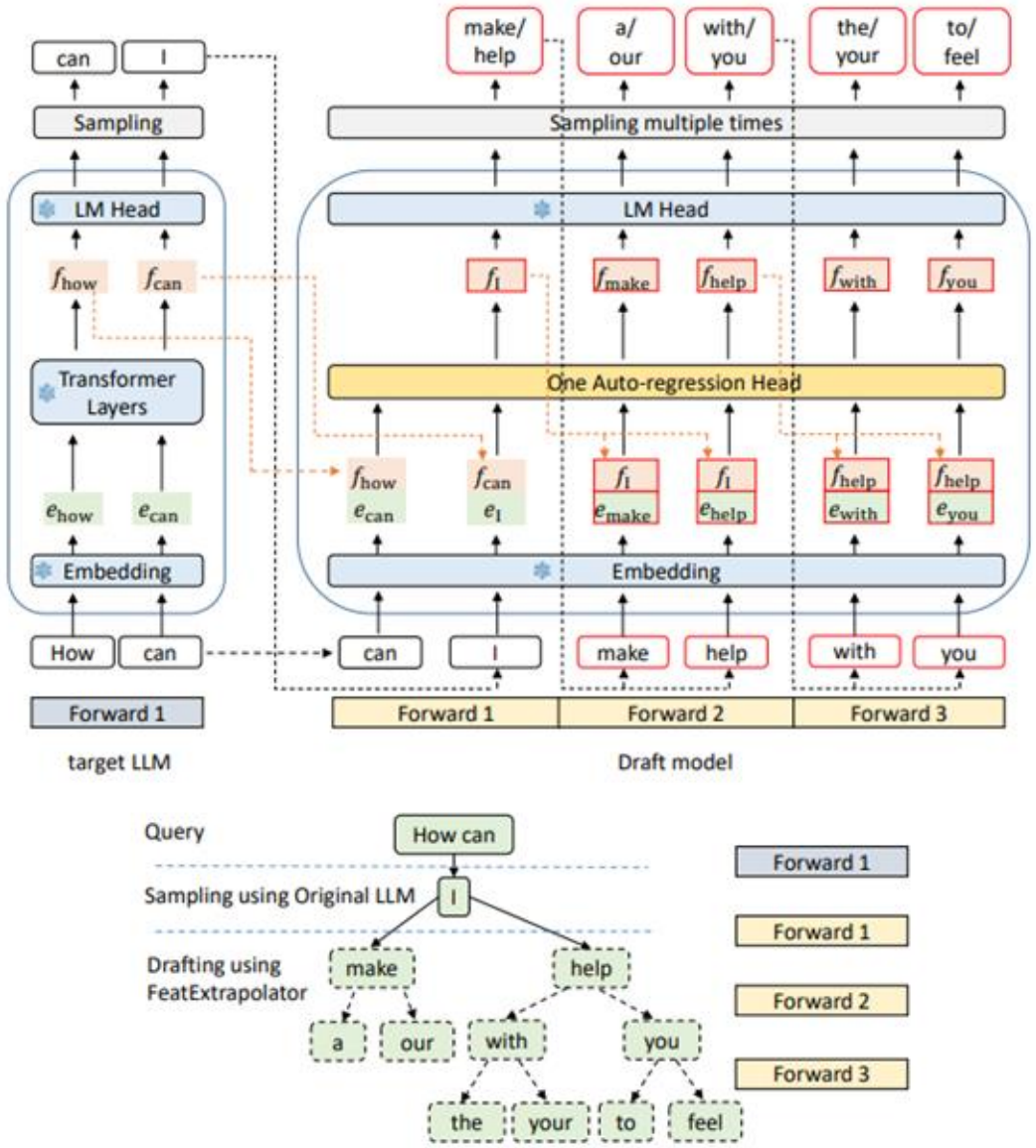
Predict future features instead of directly predicting future tokens, while using a shifted token sequence to resolve feature uncertainty

https://arxiv.org/abs/2401.15077

EAGLE (Extrapolation Algorithm for Greater Language-model Efficiency)
Speculative Sampling Requires Rethinking Feature Uncertainty

Feature-Level Speculative Decoding

- **Feature Prediction:** Predict future hidden features from the target LLM's second-to-last layer instead of directly predicting tokens.
- **Shifted Tokens:** Incorporate previously sampled tokens to reduce uncertainty in future-feature prediction.
- **Draft Tree & Verification:** Convert predicted features into candidate tokens using the target model's LM head, construct a draft tree, and verify candidates with the target model in parallel.



https://arxiv.org/abs/2401.15077

EAGLE (Extrapolation Algorithm for Greater Language-model Efficiency)
Speculative Sampling Requires Rethinking Feature Uncertainty

EAGLE Achieves ~3× Speedup

Drafting strategy	Main idea	Speedup
Token-level	Predict tokens directly	~1.9×
Feature-level	Predict future features	~1.9×
Feature + shifted tokens	Resolve feature uncertainty	**~2.8×**

Adding the shifted token sequence significantly improves feature prediction by resolving the uncertainty caused by stochastic token sampling.

https://arxiv.org/abs/2401.15077

MTP: Multi-Token Prediction
(used in DeepSeek-V2, DeepSeek-V3)

Predicting multiple future tokens through additional prediction modules.

1. Main Model

    Performs standard next-token prediction.

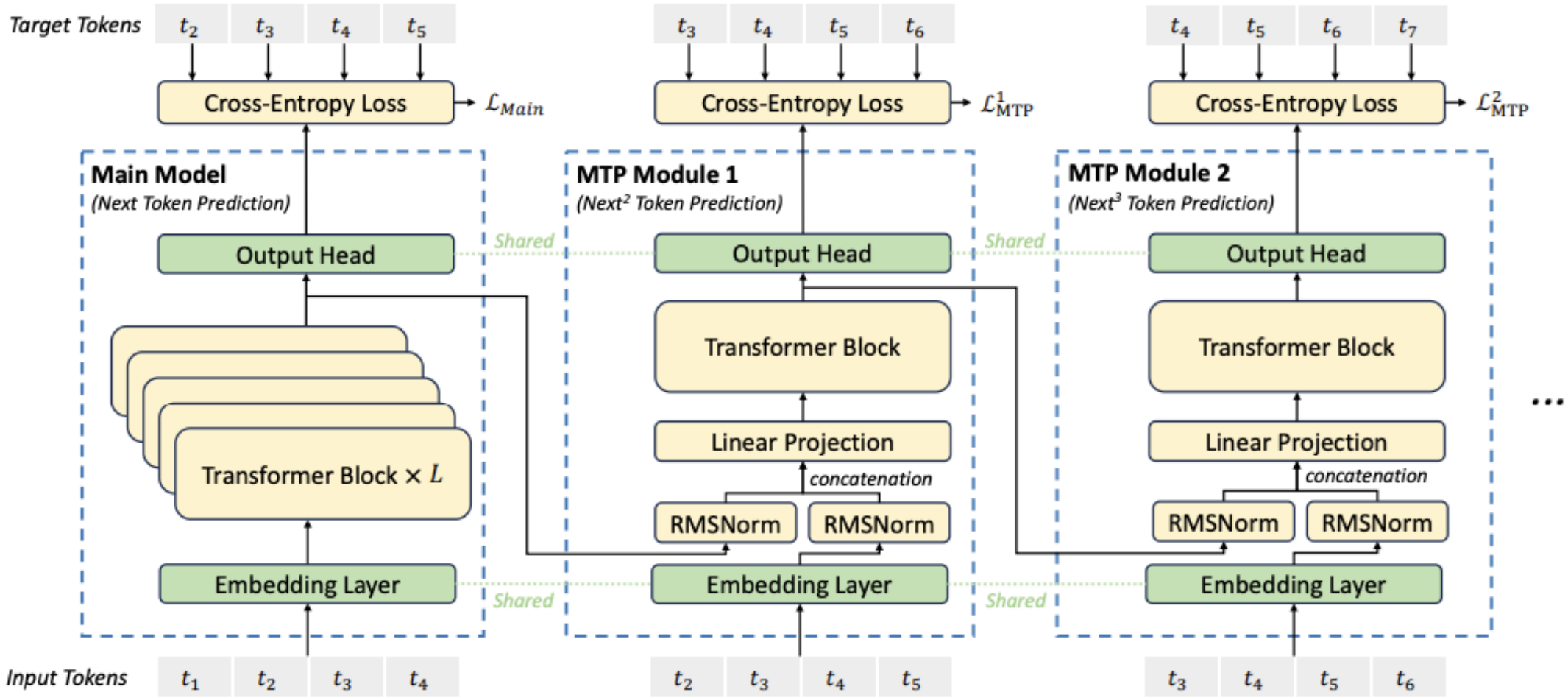
2. MTP Modules **Training needed**

    Predict additional future tokens using model representations and token information.

3. Key Advantage

    Provides additional training signals and can support faster inference with compatible implementations.

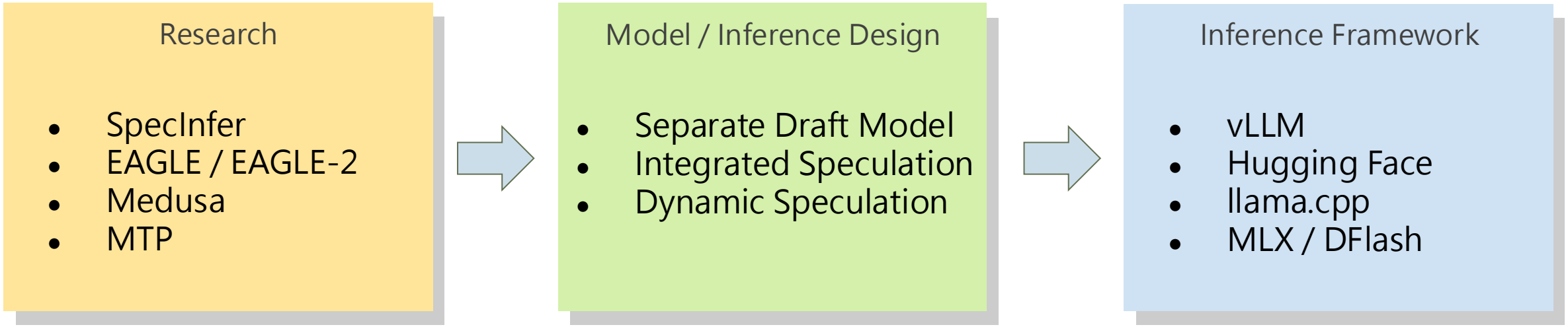
How Does MTP Work?



DeepSeek-V3

https://arxiv.org/pdf/2412.19437

From Model Architecture to Inference Framework

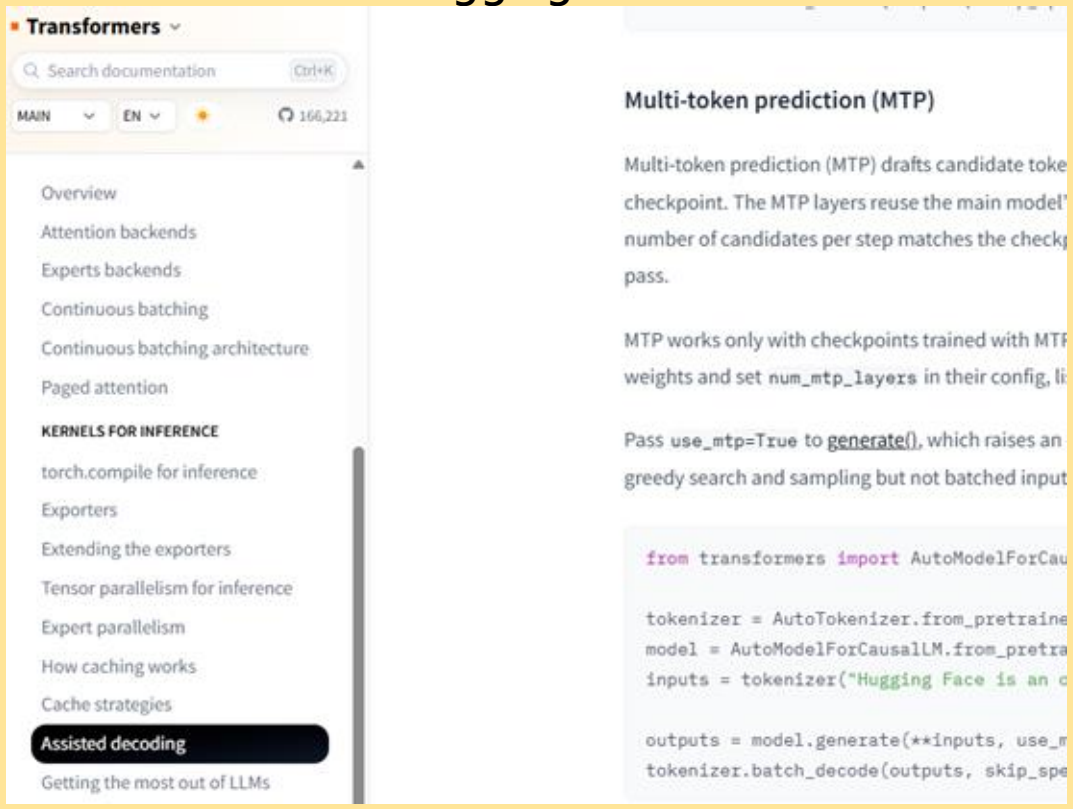


- **vLLM**：Support speculative decoding.
- **llama.cpp**：Support speculative sampling.
- **MLX + DFlash**：Do speculative inference for Apple Silicon / unified memory.

From Model Architecture to Inference Framework

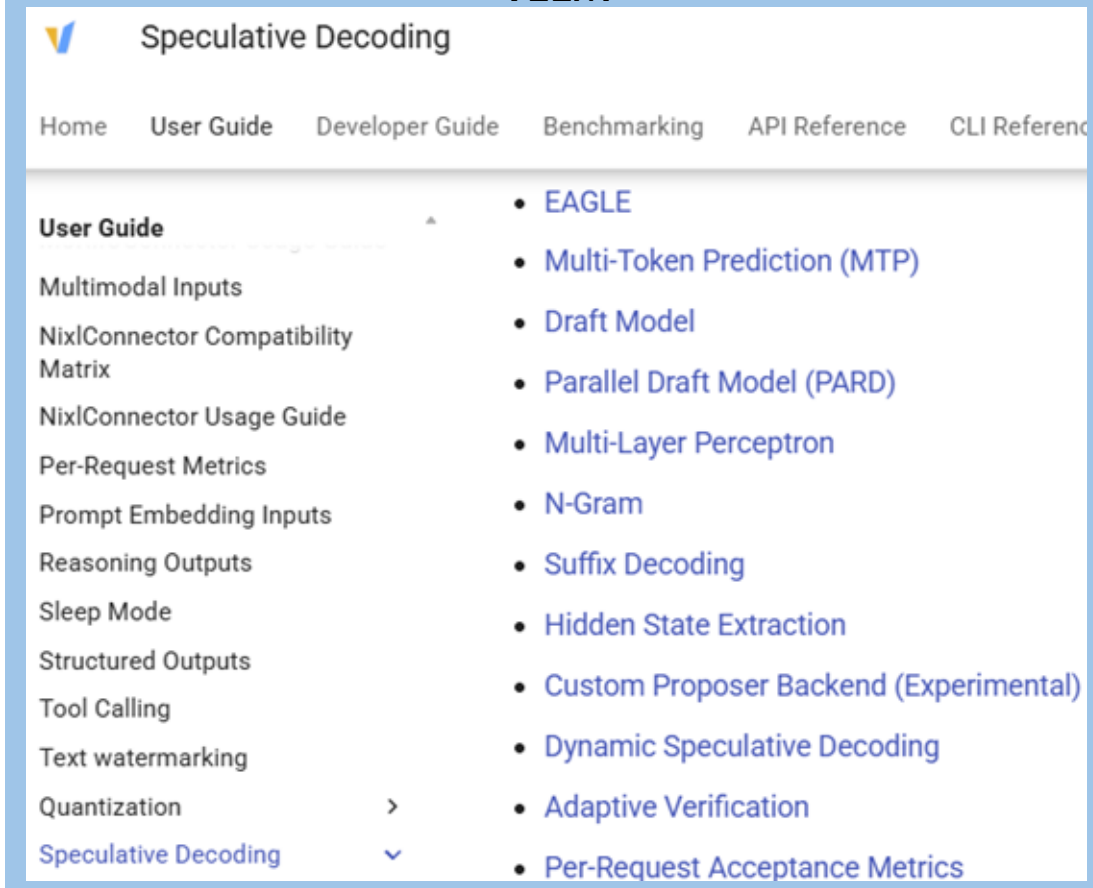
Modern frameworks such as vLLM and Hugging Face Transformers provide built-in support for multiple speculative decoding methods.

Hugging Face



https://huggingface.co/docs/transformers/main/assisted_decoding?utm_source=chatgpt.com

vLLM



https://docs.vllm.ai/en/latest/features/speculative_decoding/

Conclusion:

1. Speculative Decoding Reduces Sequential Decoding Steps

   A draft model proposes candidate tokens, and the target model verifies them in one forward pass. Multiple tokens can be accepted in a single verification cycle.

2. EAGLE and MTP Improve Multi-Token Prediction Differently

   EAGLE improves drafting through feature prediction and shifted tokens, while MTP trains additional modules to predict future tokens.

3. Practical Speedup Depends on the Entire Inference System

   Drafting cost, acceptance rate, verification overhead, hardware, and framework support determine the actual speedup.

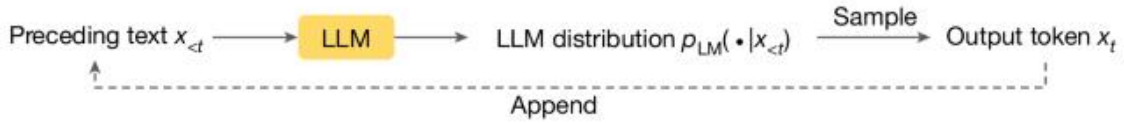
https://www.nature.com/articles/s41586-024-08025-4

Watermark for text

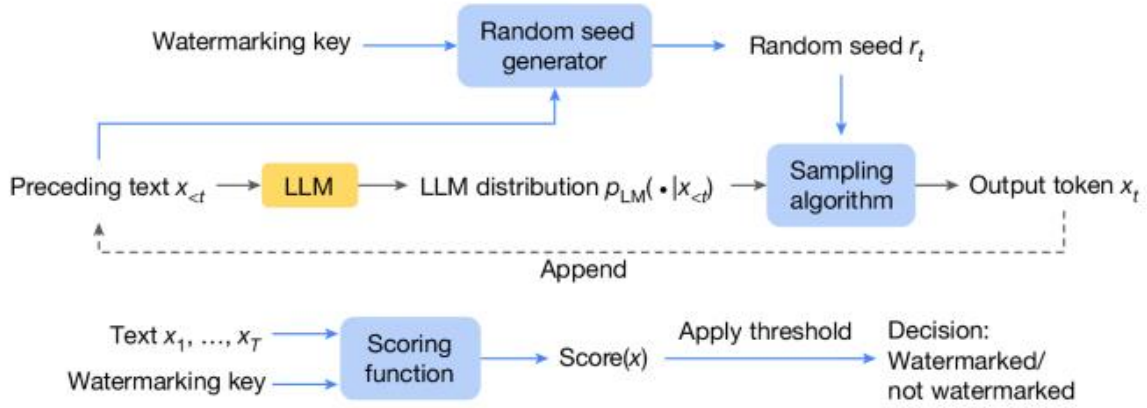
Scalable watermarking for identifying large language model outputs,

Nature 2024 Oct.

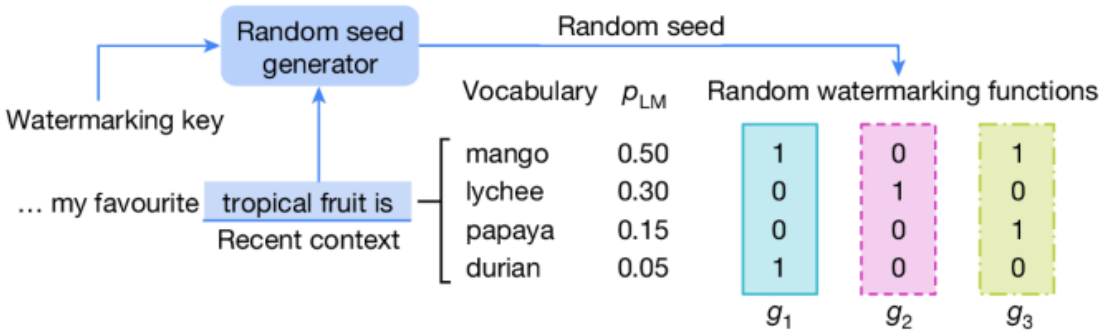
LLM text generation



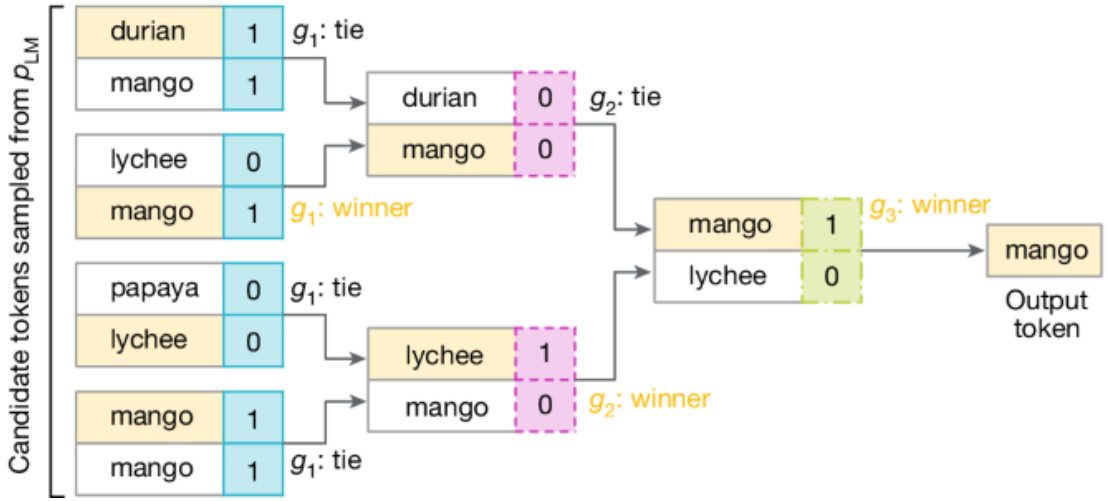
Generative watermarking: text generation and watermark detection



LLM probabilities and random watermarking functions



Tournament sampling: over-generation with watermark-based iterative selection



preserves text quality and enables high detection accuracy

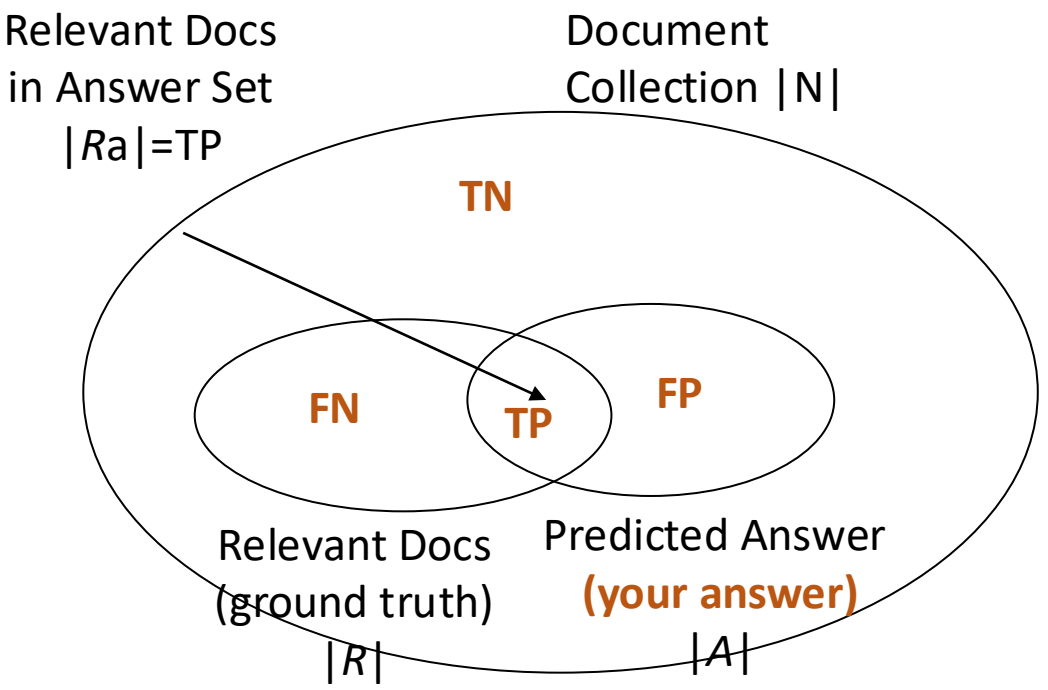
Evaluation

Alternative Measures / confusion matrix (contingency matrix?)

(predict) Answer	Relevant	
	-	+
-	TN	FN
+	FP	TP

FP: type I error, alpha error

FN: type II error, beta error



Recall (sensitivity) = |Ra| / |R| = TP / (TP + FN)

Precision = |Ra| / |A| = TP / (TP + FP)

Accuracy = (**T**N + **T**P) / |N| (For balanced domains)

classification error, *E = 1 – A*

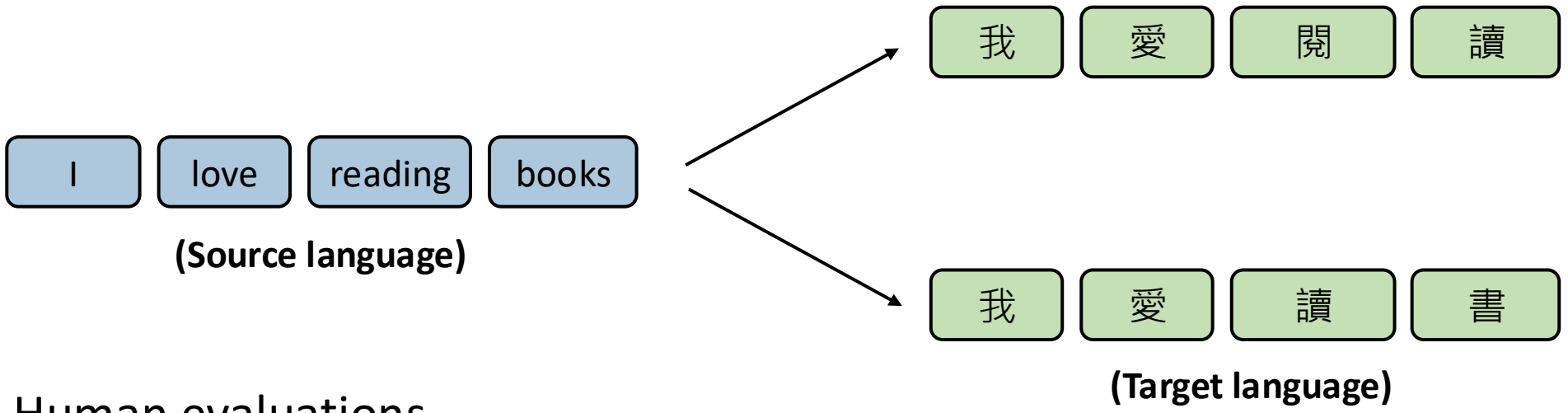
Specificity = TN / (TN + FP) (negative recall)

(not useful for Web search, TN is always so large)

How to evaluate natural language generation?

- Natural language is hard to evaluate due to subjectivity and language diversity.

For example: Machine Translation



- Human evaluations
- Automatic evaluations (We will focus on this topic.)

BLEU (Bilingual Evaluation Understudy)

- A word-based metric.
  - It is very sensitive to word tokenization
- Core concept: Compute precision for n-grams:
  - Unigrams -> BLEU-1
  - Bigrams -> BLEU-2
  - Trigrams -> BLEU-3
  - 4-grams -> BLEU-4

Papineni, Kishore, et al. "Bleu: a method for automatic evaluation of machine translation." Proceedings of the 40th annual meeting of the Association for Computational Linguistics. 2002.

Precision and Recall

$$\text{Precision} = \frac{\text{Relevant and retrieved instances}}{\text{All retrieved instances}}$$ ← Predicted by a model

$$\text{Recall} = \frac{\text{Relevant and retrieved instances}}{\text{All relevant instances}}$$ ← Ground-truths

Relevant and retrieved instances: Intersection between predictions and ground-truths

Calculation of BLEU Score (Example)

Assume we now translate from Chinese to English.

Calculate BLEU-1 score

Chinese: 我想要讀那本書

Reference1: I want to read the book.

Reference2: I want to read that book.

Model output: the the the the the the.


```
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"file_name": "000000391895.jpg",
"width": 640,
"height": 480,
"captions": [
  "A man riding a bicycle down a street next to parked cars.",
  "A person on a bike rides past cars on a city street.",
  "A man rides a bicycle past a row of parked cars.",
  "A bicyclist in a blue shirt rides by parked vehicles on the road.",
  "A person riding a bike on a street with parked cars."
]
```


An example of COCO dataset

Calculation of BLEU Score (Example)

Assume we now translate from Chinese to English.

Calculate BLEU-1 score

Chinese: 我想要讀那本書

Reference1: I want to read the book.

Reference2: I want to read that book.

Model output: the the the the the the.

Precision: $\frac{6}{6}$

100%! Can this be true?

Calculation of BLEU Score (Example)

Assume we now translate from Chinese to English.

Calculate BLEU-1 score

Chinese: 我想要讀那本書

Reference1: I want to read the book.

Reference2: I want to read that book.

Model output: the the the the the the.

~~Precision: $\frac{6}{6}$~~

Modified Precision: $\frac{1}{6}$

Papineni, Kishore, et al. "Bleu: a method for automatic evaluation of machine translation." Proceedings of the 40th annual meeting of the Association for Computational Linguistics. 2002.

Why should we use modified precision?

- The output sequences can be total mistakes.
  - E.g., the the the the the the
- Original precision is in favor of longer output sequences.
- Therefore, we should use modified precision to prevent bad evaluations.

Papineni, Kishore, et al. "Bleu: a method for automatic evaluation of machine translation." Proceedings of the 40th annual meeting of the Association for Computational Linguistics. 2002.

Calculation of BLEU Score (Example)

Calculate BLEU-2 score

Reference1: The dog is on the bed.

Reference2: There is a dog on the bed.

Model output: The dog the dog on the bed.

← More than one references can be provided for machine translation!

Papineni, Kishore, et al. "Bleu: a method for automatic evaluation of machine translation." Proceedings of the 40th annual meeting of the Association for Computational Linguistics. 2002.

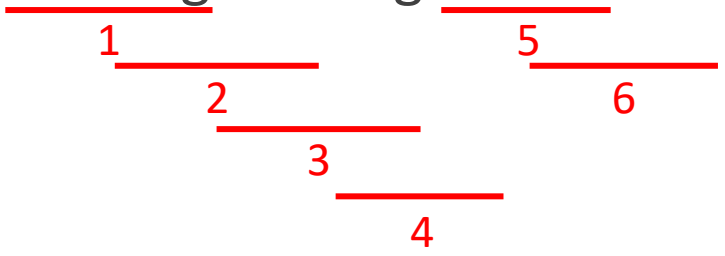
Calculation of BLEU Score (Example)

Calculate BLEU-2 score

Reference1: The dog is on the bed.

Reference2: There is a dog on the bed.

Model output: The dog the dog on the bed.



	Count	
the dog	2	(duplicated)
dog the	1	
dog on	1	
on the	1	
the bed	1	

Papineni, Kishore, et al. "Bleu: a method for automatic evaluation of machine translation." Proceedings of the 40th annual meeting of the Association for Computational Linguistics. 2002.

Calculation of BLEU Score (Example)

Calculate BLEU-2 score

Reference1: The dog is on the bed.

Reference2: There is a dog on the bed.

Model output: The dog the dog on the bed.

	Count	$\text{Count}_{\text{clip}}$ (Clips to the reference)
the dog	2	1
dog the	1	
dog on	1	
on the	1	
the bed	1	

Papineni, Kishore, et al. "Bleu: a method for automatic evaluation of machine translation." Proceedings of the 40th annual meeting of the Association for Computational Linguistics. 2002.

Calculation of BLEU Score (Example)

Calculate BLEU-2 score

Reference1: The dog is on the bed.

Reference2: There is a dog on the bed.

Model output: The dog the dog on the bed.

	Count	$Count_{clip}$
the dog	2	1
dog the	1	0
dog on	1	
on the	1	
the bed	1	

Papineni, Kishore, et al. "Bleu: a method for automatic evaluation of machine translation." Proceedings of the 40th annual meeting of the Association for Computational Linguistics. 2002.

Calculation of BLEU Score (Example)

Calculate BLEU-2 score

Reference1: The dog is on the bed.

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	Count	$Count_{clip}$
the dog	2	1
dog the	1	0
dog on	1	1
on the	1	
the bed	1	

Papineni, Kishore, et al. "Bleu: a method for automatic evaluation of machine translation." Proceedings of the 40th annual meeting of the Association for Computational Linguistics. 2002.

Calculation of BLEU Score (Example)

Calculate BLEU-2 score

Reference1: The dog is on the bed.

Reference2: There is a dog on the bed.

Model output: The dog the dog on the bed.

Count only one time even mapped to both references.

	Count	$Count_{clip}$
the dog	2	1
dog the	1	0
dog on	1	1
on the	1	1
the bed	1	

Papineni, Kishore, et al. "Bleu: a method for automatic evaluation of machine translation." Proceedings of the 40th annual meeting of the Association for Computational Linguistics. 2002.

Calculation of BLEU Score (Example)

Calculate BLEU-2 score

For each unique ngram
- count its maximum frequency in each of the reference sentences.
- The clipped count = minimum (this special count, the original count)

Reference1: The dog is on the bed.

Reference2: There is a dog on the bed.

Model output: The dog the dog on the bed.

Count only one time even mapped to both references.

	Count	$\text{Count}_{\text{clip}}$
the dog	2	1
dog the	1	0
dog on	1	1
on the	1	1
the bed	1	1

Modified Precision: $\frac{4}{6}$

Papineni, Kishore, et al. "Bleu: a method for automatic evaluation of machine translation." Proceedings of the 40th annual meeting of the Association for Computational Linguistics. 2002.

Formula of BLEU Score

Summation for unigram, bigram, tri-gram, and 4-gram

$$p_n = \frac{\sum_{C \in \{Candidates\}} \sum_{n\text{-}gram \in C} Count_{clip}(n\text{-}gram)}{\sum_{C' \in \{Candidates\}} \sum_{n\text{-}gram' \in C'} Count(n\text{-}gram')}$$

Summation for all candidates (model outputs) of each translation

Papineni, Kishore, et al. "Bleu: a method for automatic evaluation of machine translation." Proceedings of the 40th annual meeting of the Association for Computational Linguistics. 2002.

What we've learned BLEU so far

- The BLEU score is calculated from the summation of 1-gram to 4-gram.
  - You can also measure n-gram individually.
- We use modified precision to prevent bad evaluations.
- What will happen if a model tends to generate really short sentences?

➡ **More penalty for calculating BLEU score!**

Papineni, Kishore, et al. "Bleu: a method for automatic evaluation of machine translation." Proceedings of the 40th annual meeting of the Association for Computational Linguistics. 2002.

Brevity Penalty (BP)

- BP is used to penalize short candidates.

c: The length of a candidate sequence
r: The length of a reference sequence that is closest to c (shorter one)

$$\mathrm{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

Then,

$$\mathrm{BLEU} = \mathrm{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

N=4 to include 1-gram to 4-gram

Weight for each n-gram (was set 1/4 in the original paper)

Papineni, Kishore, et al. "Bleu: a method for automatic evaluation of machine translation." Proceedings of the 40th annual meeting of the Association for Computational Linguistics. 2002.

https://aclanthology.org/W04-1013.pdf

ROUGE: A Package for Automatic Evaluation of Summaries (Recall-Oriented Understudy for Gisting Evaluation)

- ROUGE-N: N-gram Co-Occurrence Statistics (recall base)
- ROUGE-L: count LCS

$$\text{ROUGE-N} = \frac{\sum_{S \in \{ReferemceSummaries\}} \sum_{gram_n \in S} Count_{match}(gram_n)}{\sum_{S \in \{ReferenceSummaries\}} \sum_{gram_n \in S} Count(gram_n)}$$

S1. police killed the gunman (reference)
S2. police kill the gunman (summary candidate 1)
S3. the gunman kill police (summary candidate 2)

$ROUGE\text{-}2_{S2} = ROUGE\text{-}2_{S3}$

$ROUGE\text{-}L_{S2} = ¾ > ROUGE\text{-}L_{S3} = ½$

police the gunman vs the gunman

S4. the gunman police killed

$ROUGE\text{-}2_{S4} > ROUGE\text{-}2_{S2}$

$ROUGE\text{-}L_{S4} < ROUGE\text{-}L_{S2}$

(Recap) Perplexity

Perplexity (PPL) is a quantitative criterion used to evaluate the capacities of language modeling models.

- Given the sequence of words $W = w_1 w_2 \dots w_N$ and an N-gram model. The PPL of the model was computed by:

$$Perplexity(W) = P(w_1 w_2 \dots w_N)^{-\frac{1}{N}} = \sqrt[N]{\prod_{k=1}^{n} \frac{1}{P(w_k | w_{k-N+1:k})}}$$

The lower the value of perplexity, the better the language modeling capability of the model.

Comparison for Human and Automatic Evaluations

- Human evaluations
  - Pros: More accurate for subjectivity, flexibility for any desired comparison
  - Cons: Less objective, time-consuming, expensive
- Automatic evaluations
  - Pros: Objective enough to serve as common evaluation metrics, fast
  - Cons: Cannot meet language diversity
    - Take machine translation for instance, there are always other valid ways to translate the source sentence.

NLP Benchmark – GLUE ICLR 2019, https://openreview.net/pdf?id=rJ4km2R5t7

GLUE: A MULTI-TASK BENCHMARK AND ANALYSIS PLATFORM FOR NATURAL LANGUAGE UNDERSTANDING

GLUE: General Language Understanding Evaluation

- Consist of **9** task
  - Using one model to fit all tasks
  - Prove that model have **general language understanding**
- Task can be divided into **three** categories
  - **Single** Sentence Classification
  - **Pairwise** Text Classification
  - Text **Similarity**
- Average score
  - Average 9 task performance
  - **Higher** means better
  - **Human** baseline: **87.6** on average

NLP Benchmark - GLUE

Single Sentence Classification

- Sentence **Acceptability**
  - Given a sentence **s**, is **s** a valid sentence, i.e., have **correct grammar** and **meaningful content**?
  - Dataset: **CoLA**
- **Sentiment** Analysis
  - Given a sentence **s**, is **s** expressing **postive** sentiment or **negative** sentiment?
  - Dataset: **SST-2**

Pairwise Text Classification

- Natural Language **Inference** (NLI)
  - Given premise **p** and hypothesis **h**, is p **implies**, **contradict** or **not related** to **h**?
  - Dataset: **MNLI-(m/mm)**, **RTE**, **WNLI**, **QNLI**
- Paraphrase
  - Given two sentences **s1** and **s2**, is **s1 means s2**, i.e., **s1** is "**in other words**" **s2**?
  - Dataset: **QQP**, **MRPC**

Text **Similarity**

- Given two sentences **s1** and **s2**, from **1 to 10**, how much does **s1 similar to s2**?
- Dataset: **STS-B**

GLUE Benchmark

https://paperswithcode.com/dataset/glue

Data source: https://gluebenchmark.com/tasks

Data Name	Full Name	Train / Val Size	Task
CoLA	The Corpus of Linguistic Acceptability	8.6k / 1.0k / 1.1k	Single-sentence Classification
SST-2	The Stanford Sentiment Treebank	67k / 87k / 1.8k	Single-sentence Classification
MRPC	Microsoft Research Paraphrase Corpus	3.7k / 408 /1.7k	Pair-sentence Classification
STS-B	Semantic Textual Similarity Benchmark	5.8k / 1.5k / 1.4k	Semantic Similarity
QQP	Quora Question Pairs	364k / 40.4k / 391k	Question Answering
MNLI-m	Multi-Genre Natural Language Inference (matched)	393k / 9.8k / 9.8k	Pair-sentence Classification
MNLI-mm	Multi-Genre Natural Language Inference (mis-matched)	393k / 9.8k / 9.9k (Same training set as MNLI-m)	Pair-sentence Classification
QNLI	Question NLI	105k / 5.5k / 5.5k	Pair-sentence Classification
RTE	Recognizing Textual Entailment	2.5k / 277 / 3k	Pair-sentence Classification
WNLI	Winograd NLI	635 / 71 / 146	Pair-sentence Classification

https://nyu-mll.github.io/CoLA/

CoLA: The Corpus of Linguistic Acceptability

imbalanced dataset (7:3)

Corpus Sample

clc95	0	*	In which way is Sandy very anxious to see if the students will be able to solve the homework problem?
c-05	1		The book was written by John.
c-05	0	*	Books were sent to each other by the students.
swb04	1		She voted for herself.
swb04	1		I saw that gas can explode.

"The students sent books to each other."

CoLA hard examples

X I can't go because too tired.

X The child seems sleeping.

O The pizza cried all night.

https://rajpurkar.github.io/SQuAD-explorer/

NLP Benchmark - SQuAD

SQuAD: Stanford Question Answering Dataset

- A **reading comprehension** dataset
- Consisting of **questions** posed by crowdworkers on a set of **Wikipedia** atricles
- The **answer** to every question is **a segment of text from the corresponding reading passage**
- Or the question might be **unanswerable**

SQuAD 1.1

- 100000+ question-answer pairs
- 500+ articles

SQuAD 2.0

- SQuAD 1.1 + 50000 **unanswerable** questions
- Unanswerable questions look similar to answerable ones.
- To do well on SQuAD 2.0, systems must not only answer questions when possible, but also determine when no answer is supported by the paragraph and abstain from answering

SQuAD 2.0

Computational_complexity_theory

The Stanford Question Answering Dataset

Computational complexity theory is a branch of the theory of computation in theoretical computer science that focuses on classifying computational problems according to their inherent difficulty, and relating those classes to each other. A computational problem is understood to be a task that is in principle amenable to being solved by a computer, which is equivalent to stating that the problem may be solved by mechanical application of mathematical steps, such as an algorithm.

What branch of theoretical computer science deals with broadly classifying computational problems by difficulty and class of relationship?
Ground Truth Answers: Computational complexity theory | Computational complexity theory | Computational complexity theory

By what main attribute are computational problems classified utilizing computational complexity theory?
Ground Truth Answers: inherent difficulty | their inherent difficulty | inherent difficulty

Leaderboard

SQuAD2.0 tests the ability of a system to not only answer reading comprehension questions, but also abstain when presented with a question that cannot be answered based on the provided paragraph.

Rank	Model	EM	F1
	Human Performance *Stanford University* (Rajpurkar & Jia et al. '18)	86.831	89.452
1 Jun 04, 2021	IE-Net (ensemble) *RICOH_SRCB_DML*	**90.939**	**93.214**

New QA Benchmarks

長文本跨章節檢索與推理

- Needle In A Haystack、L-Eval、InfiniteBench、RULER。

多跳深度推理（Multi-hop Reasoning QA）

- HotpotQA、MuSiQue。

專家級學術與科學問答

- GPQA（Diamond）、PubMedQA、HLE（Humanity's Last Exam）。

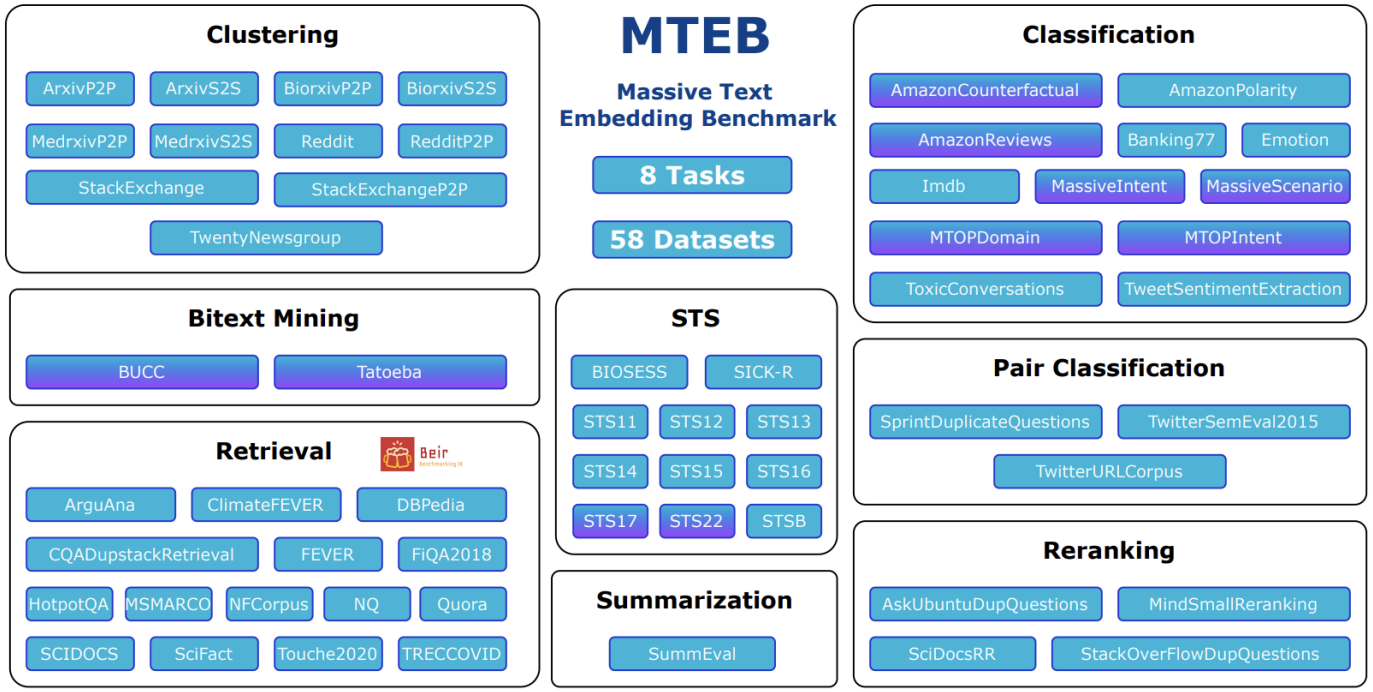
https://arxiv.org/abs/2210.07316

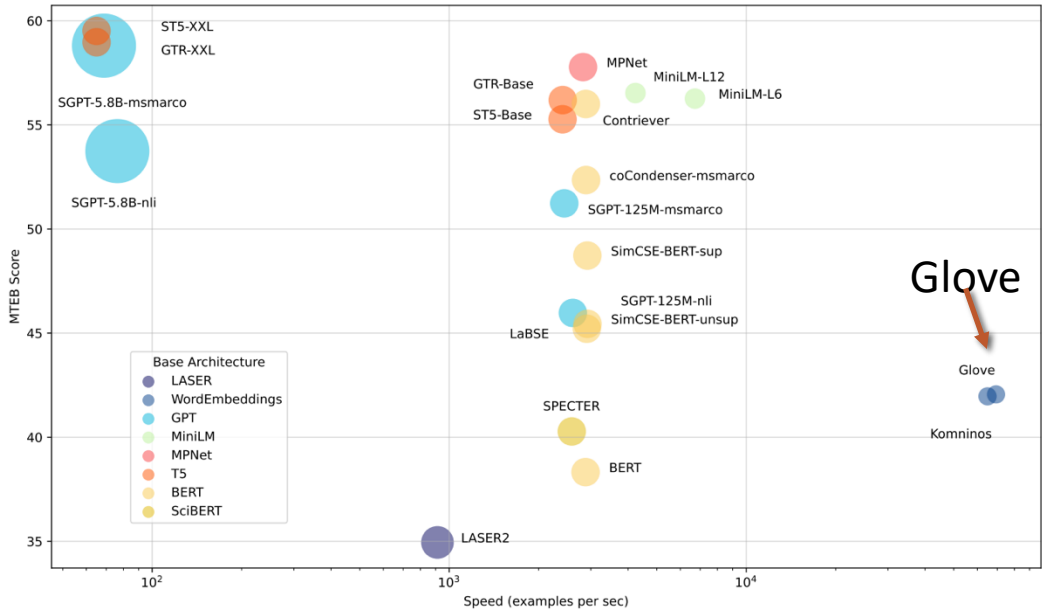
https://github.com/embeddings-benchmark/mteb

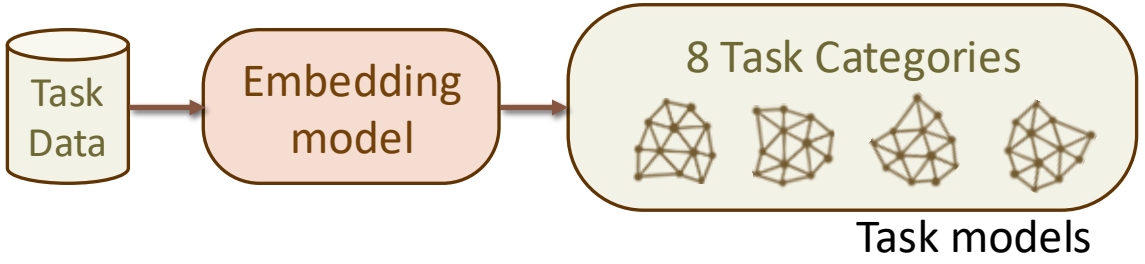
MTEB (Massive Text Embedding Benchmark)

(mMTEB, C-MTEB)

https://huggingface.co/spaces/mteb/leaderboard







SOTA: Qwen-Embedding-8B (~~70.58~~)

(nDCG@10, Recall@k, Accuracy, F1, V-measure, ARI, Spearman, Pearson correlation)

In-domain, out-of-domain, and open-domain data

In-Domain Data

•**Definition**: In-domain data refers to data that is **similar** to the data used to train a model. It typically comes from the same distribution, context, or domain. (Aligned with the training data)

•**Examples**: If a model is trained on news articles about technology, in-domain data would also consist of technology-related news articles.

Out-of-Domain Data

•**Definition**: Out-of-domain data consists of data that is **significantly different** from the training data, either in terms of content, style, or context.

•**Examples**: Continuing with the previous example, if the model is tested on health-related articles, this would be considered out-of-domain data.

Open-Domain Data

•**Definition**: Open-domain data refers to data that covers a wide range of topics and is not restricted to a specific domain. It is designed to be general and comprehensive.

•**Examples**: A chatbot that can answer questions about history, science, sports, and technology would work with open-domain data.

https://paperswithcode.com/dataset/mmlu

MMLU: Massive Multitask Language Understanding

1 measure knowledge acquired during pretraining by evaluating models exclusively in **zero-shot** and **few-shot** settings.

2 covers 57 subjects across STEM, the humanities, the social sciences, and more.

3 multiple-choice QA

Also TMMLU, MediaTek
https://arxiv.org/pdf/2309.08448

Also TMMLU+, iKala (TMMLUx6)
https://arxiv.org/abs/2403.01858

Q: 「臺灣原住民的布只有形制屬傳統或較現代的分別，像圓領的剪裁、鈕扣和棉布的使用等，都是受漢人的影響而來。泰雅族的貝珠鈴衣，是貝珠串底下加銅鈴裝飾，銅鈴也是和漢人交易而來。日治時代的原住民服裝，還出現以漢人棉布做底、日本布做袖口、原住民圖案做主要裝飾的混搭法。」這段文字的主旨最可能是下列何者？
(A)不同文化的碰撞，可融合並產生新的火花
(B)外來文化的入侵，讓在地的傳統文化日漸消失
(C)臺灣原住民的文化，影響了漢人與日本人的穿著
(D)觀察不同族群的服飾，就能了解不同文化的差異

A: (A)